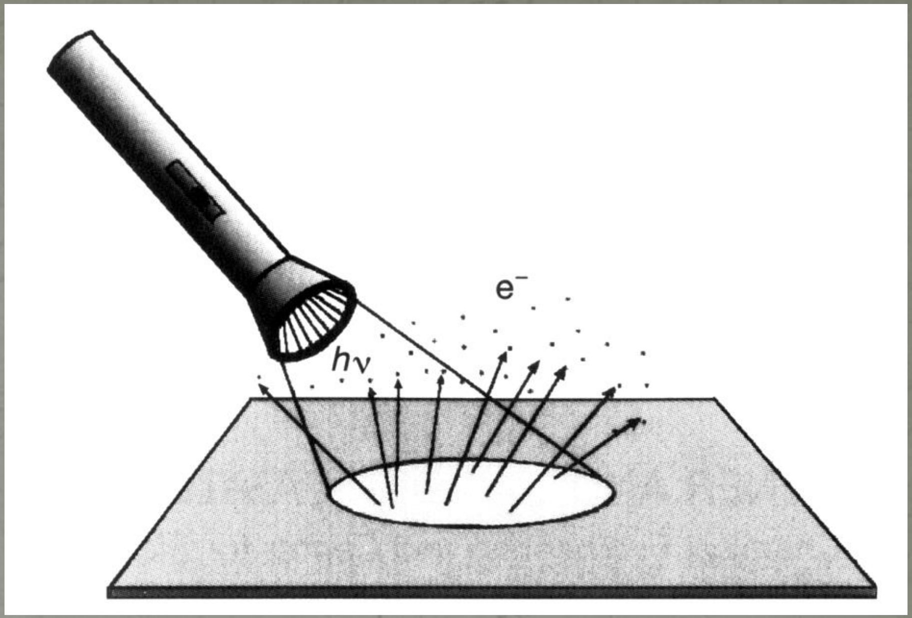
X-RAY PHOTOELECTRON SPECTROSCOPY (XPS)

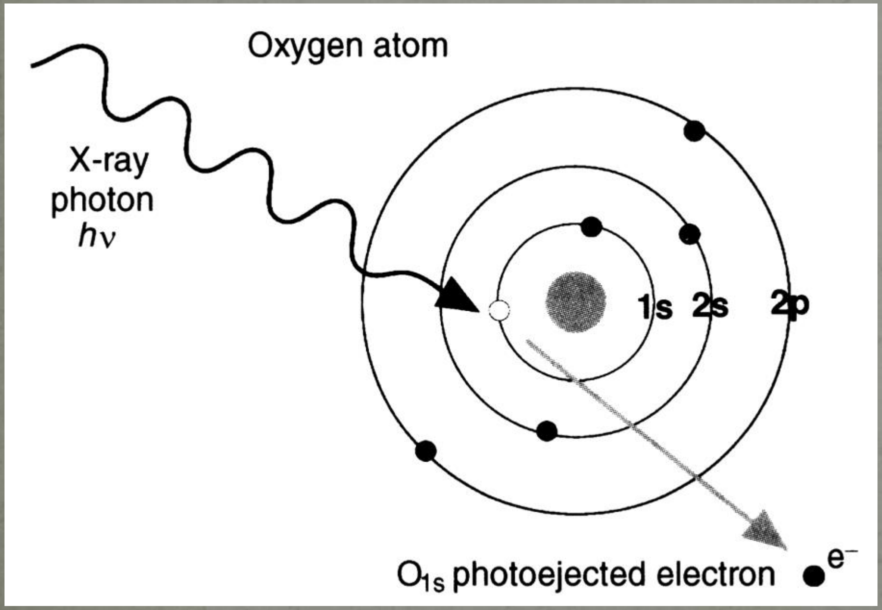
Introduction to xps

- What is XPS? General Theory
- How can we identify elements and compounds?
- What is the instrumentation of XPS?

Photoelectric Effect



Einstein, Nobel Prize 1921



Photoemission as an analytical tool

Kai Siegbahn, Nobel Prize 1981

Ångströmlaboratoriet

WHAT IS XPS?

- *X-ray Photoelectron Spectroscopy (XPS) is also known as Electron Spectroscopy for Chemical Analysis (ESCA) is a widely used technique to investigate the chemical composition of surfaces.*

  - Measure the elemental composition, chemical stoichiometry, chemical state, and electronic state of the elements that exist in a material

Background of xps

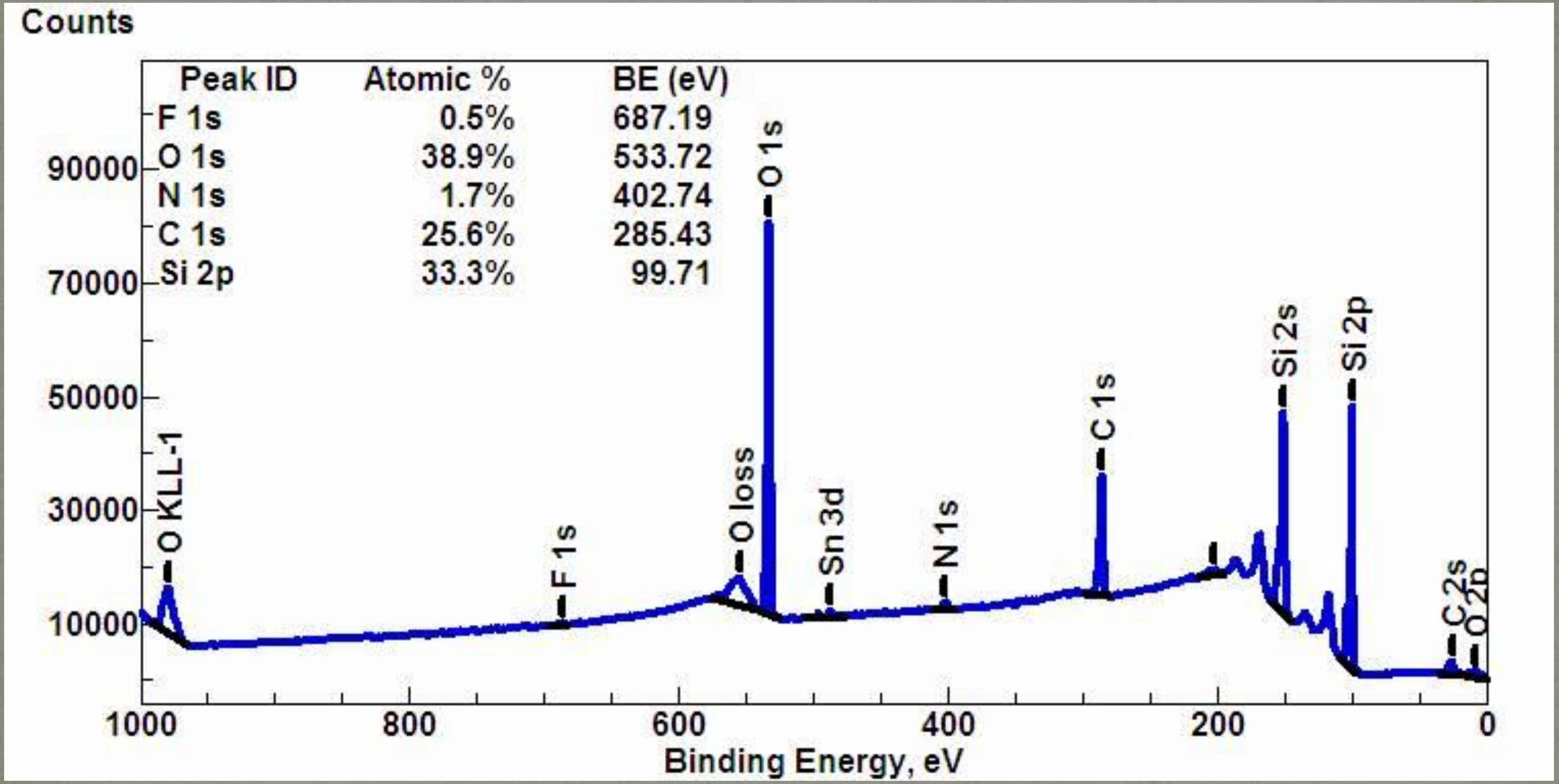
- 1.In 1887, Henriech Rudolf Hertz,photoelectric effect

  2.In 1907, Innes discovered the first XPS spectrum

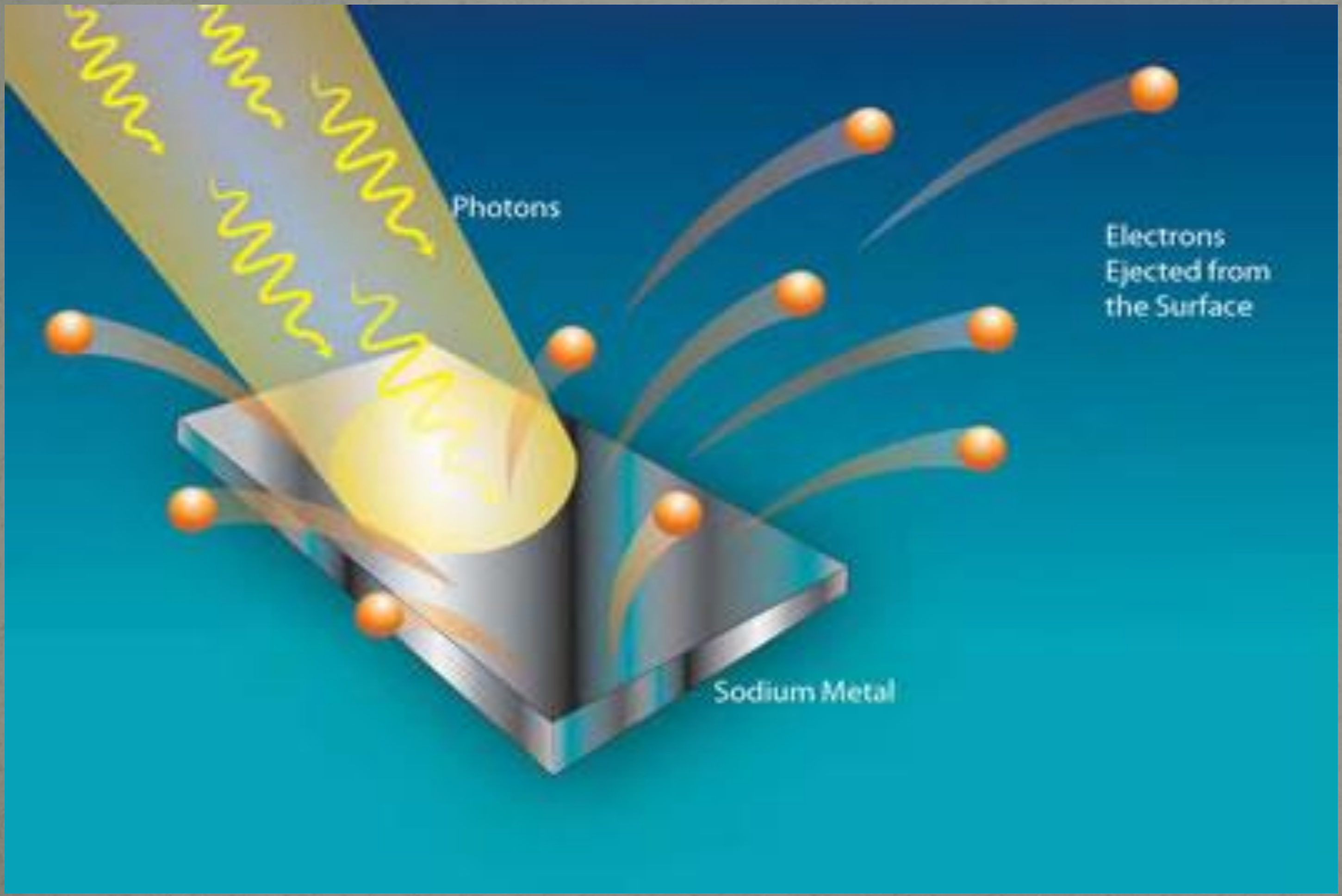
  3.Kai Siegbahn(Uppsala) with Hewlett Packard

  (USA) produce the first commertial monochromatic XPS

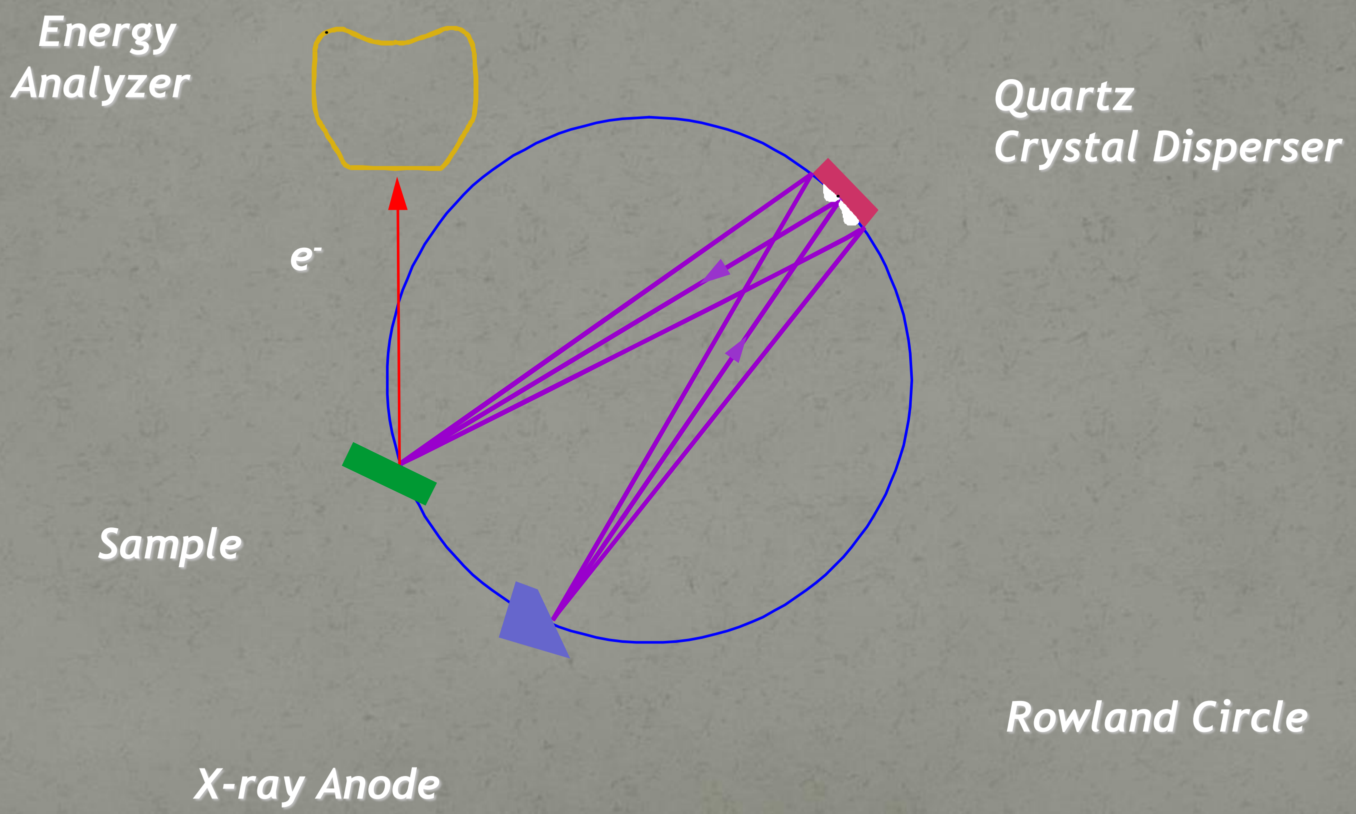
Wide-scan or survey spectrum of a somewhat dirty silicon wafer, showing all elements present.



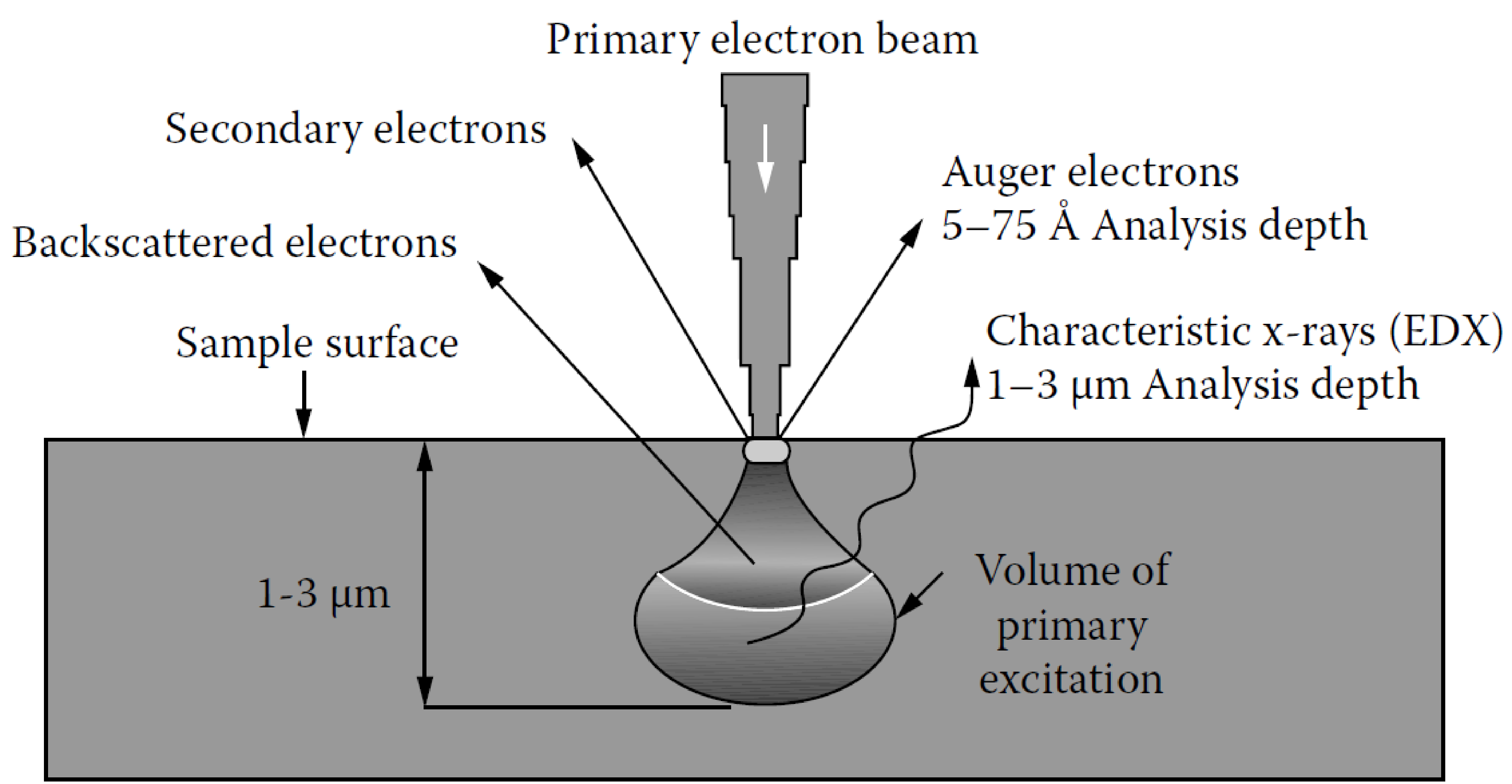
Photoelectrict effect



Schematic of X-ray Monochromator



Generation of electron beam

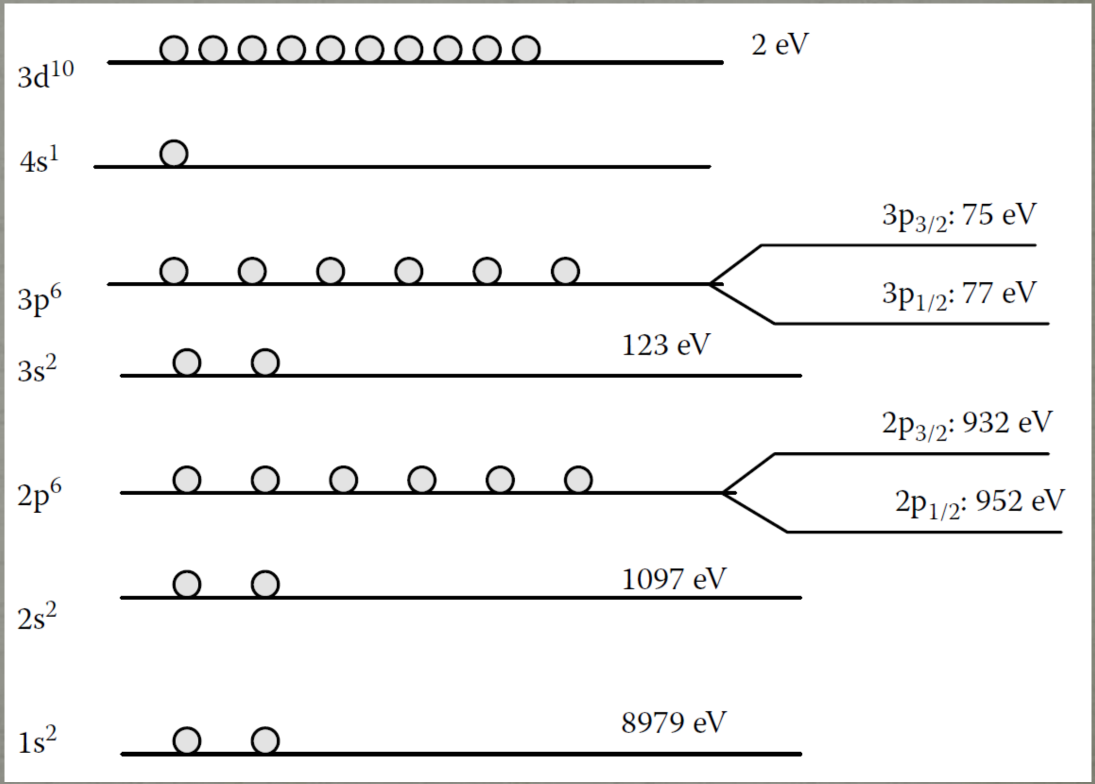


Atomic Model and Electron Configuration

Maximum Number of Electrons the Orbital Can Accommodate				
Orbit	*s*	*p*	*d*	*f*
Maximum no. of electrons	2	6	10	14

The electron placement in the main shells (the primary quantum numbers) and subshells (the orbital) according to the Aufbau principle is as follows:

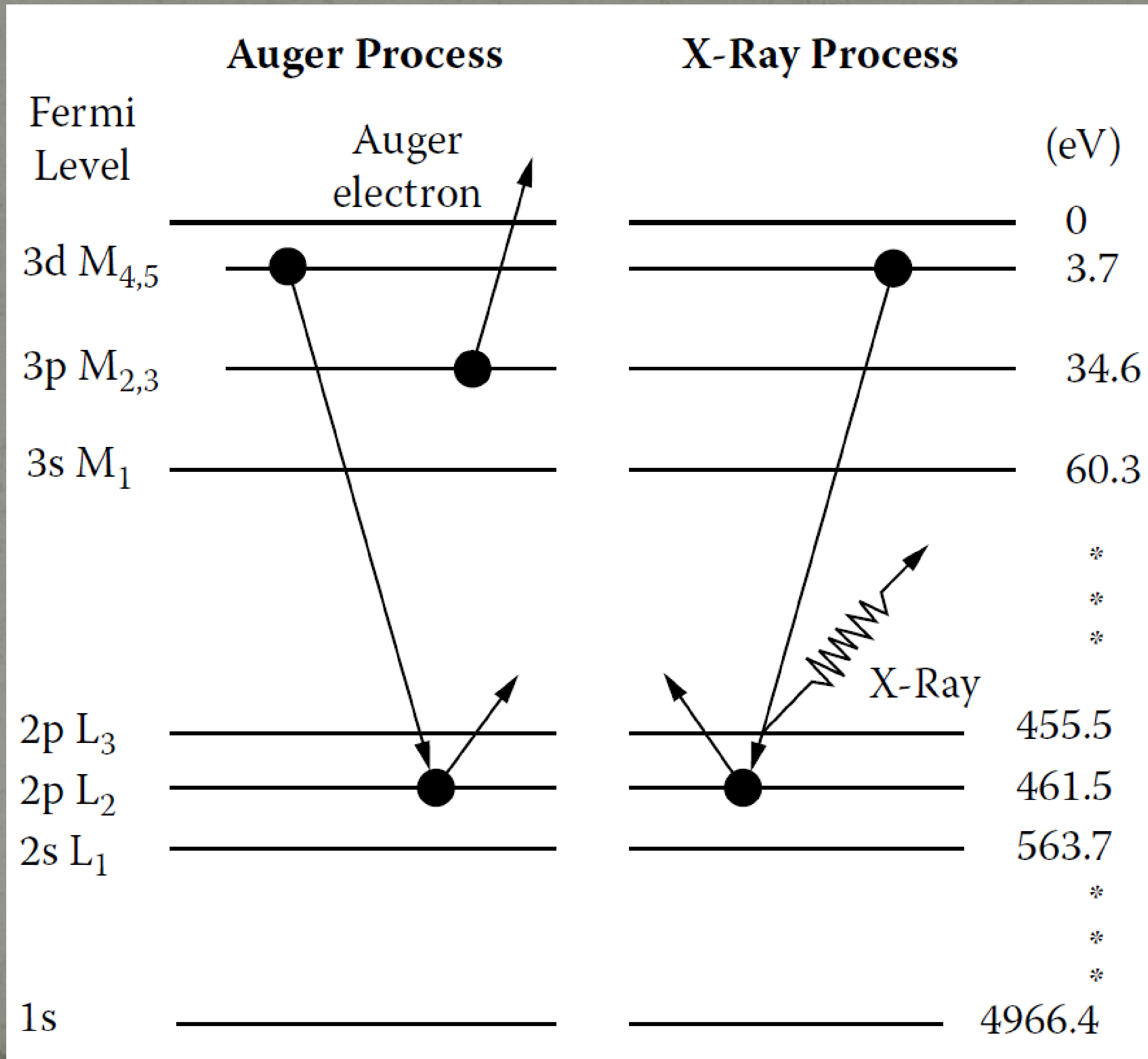
1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f 5d 6p 7s 5f 6d 7p.



Electron configuration of Cu and the binding energies

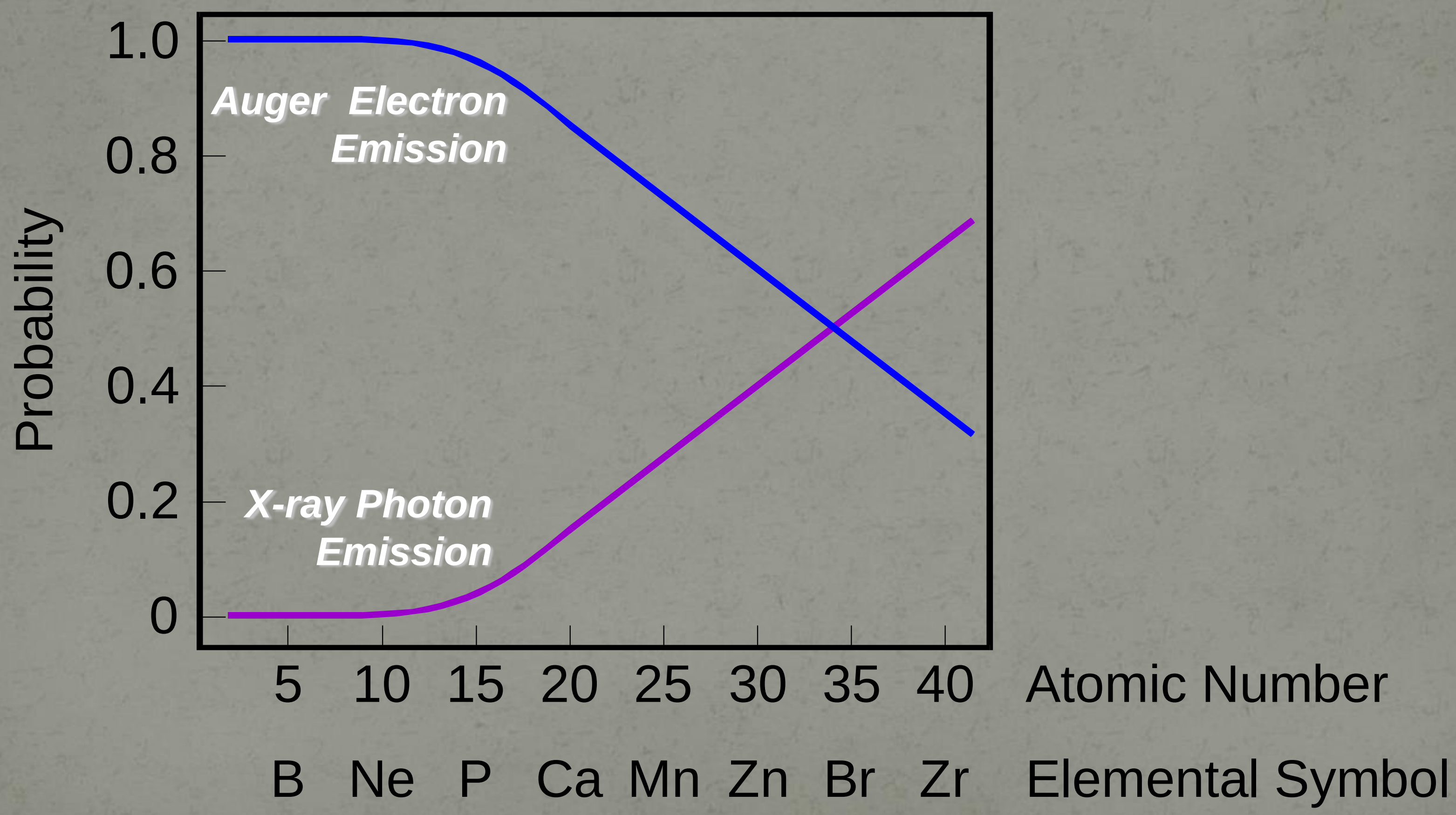
$_{29}Cu$: $1s_2\ 2s_2\ 2p_6\ 3s_2\ 3p_6\ 4s_1\ 3d_{10}$

Auger and x-ray processes after ionization



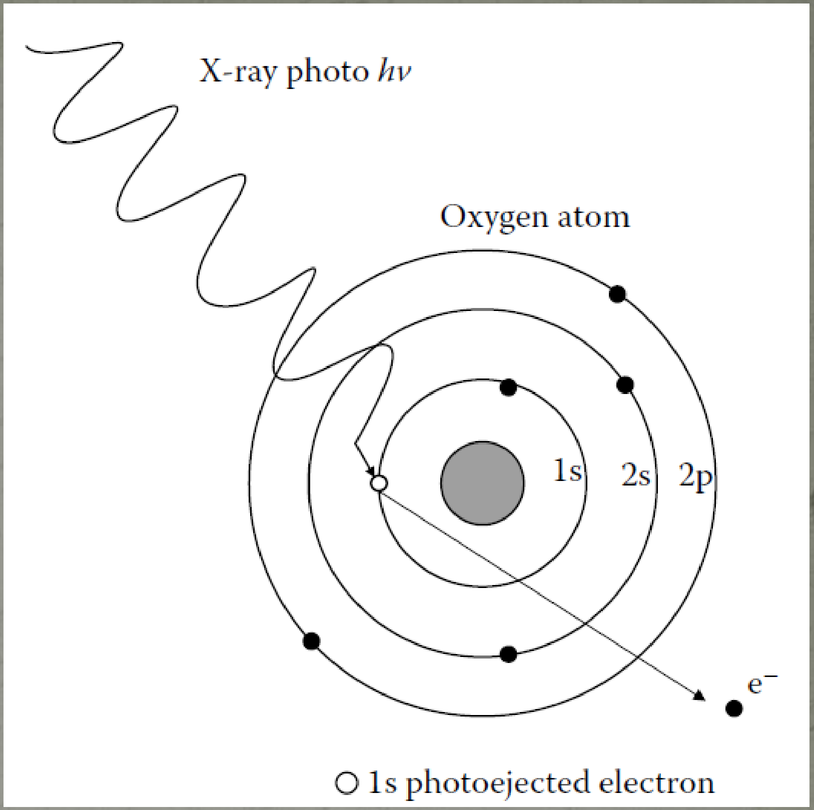
Homework

Relative Probabilities of Relaxation of a K Shell Core Hole



Photoionization

Photoionization is the physical process in which an incident x-ray photon with energy of hv ejects one or more electrons from an atom, ion, or molecule.



- Not every photon that encounters an atom or ion will photoionize it. The probability of photoionization is related to the photoionization cross section, which depends on the energy of the photon and the target element being considered.

- For photon energies below the ionization threshold, the photoionization cross section is zero. Above the threshold, the cross section decreases as E^{-3} (photon energy)

The process of photoionization can be written as:

$$A + h\nu \rightarrow A^{+} + e^{-}.$$

The left-hand side is an atom of no charge (neutral), and a photon of energy *hv*; the right-hand side is an ionized atom (now one positive charge) plus an ejected electron (one negative charge).

From the energy viewpoint, the energies before and after the striking of the photon on the nucleus should be equal (conservation of energy):

$$E(\mathrm{A}) + h\nu = E(\mathrm{A}^{+}) + E(\mathrm{e}^{-}).$$

Since the energy of the ejected electron E(e^-) is present solely as kinetic energy (E_K)

$$E_K = h\nu - \underline{[E(A^{+}) - E(A)]}.$$

This term is actually the difference in energy between the ionized and neutral atoms. In fact, that is what we call the binding energy (E_B) of the electron.

Therefore, above Equation can be rewritten as

$$E_K = h\nu - E_B.$$

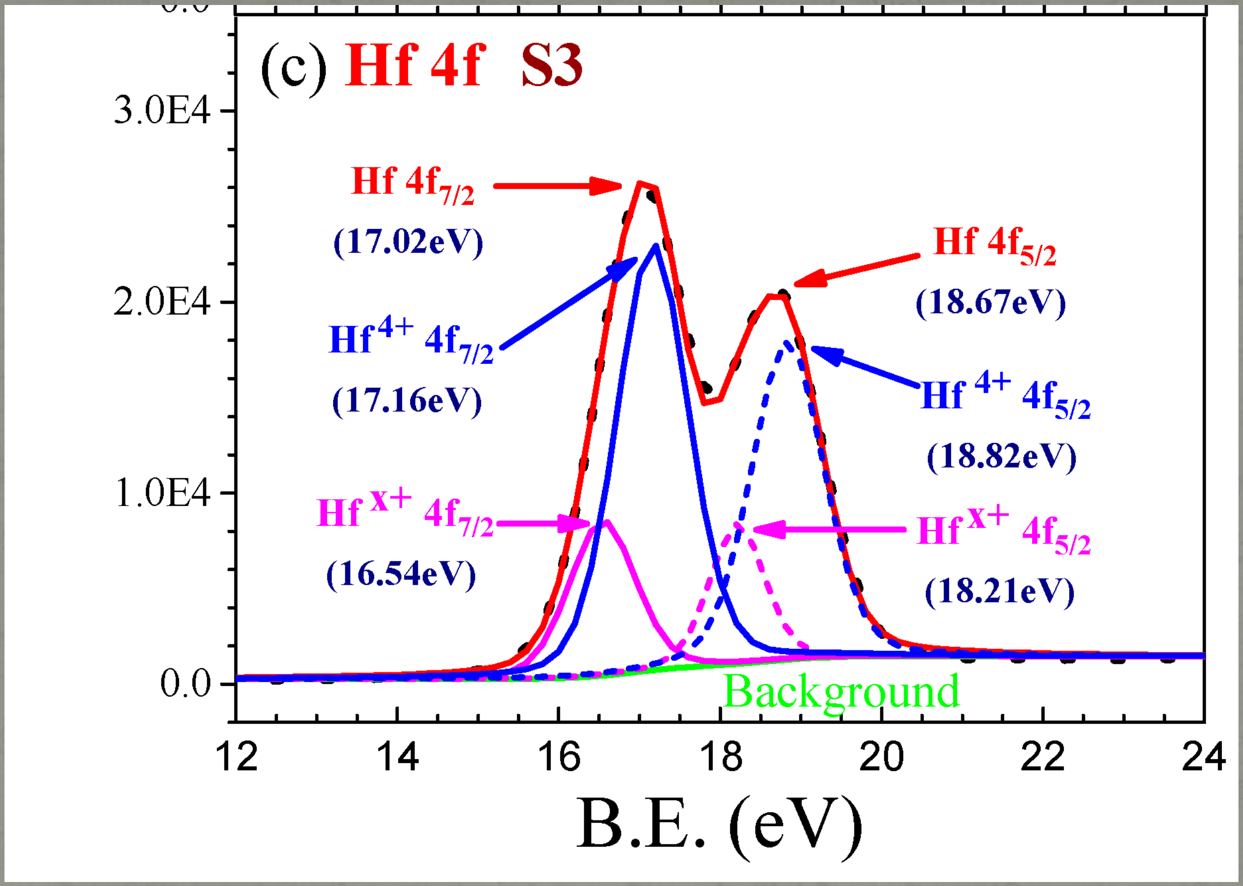
E_B is a direct measure of the energy required to remove only the electron concerned from its initial level to the vacuum level.

- E_B is unique for a given atom and a given orbital.
- The kinetic energy E_K of the photoelectron is the energy of the ejected electron after it leaves the “hold” of the atom. E_K varies when a different photon energy (*hv*) of the x-ray is used E_K is therefore not an intrinsic material property
- The E_B of the electron is that which identifies the electron specifically, both in terms of its parent element and atomic energy level.
- As E_K is measurable for a given x-ray source, *hv* is known. Thus it is easy to calculate E_B, from which one can identify the atom (element) that a particular electron had escaped from because no two elements share the same set of electronic binding energies.
- At a given photon energy, any changes in E_B are reflected in E_K, which means that changes in the chemical environment of an atom can be followed by monitoring the changes in the photoelectron energies, leading to the provision of chemical information.

Elemental Shifts

	Binding Energy (eV)		
Element	***2p$_{3/2}$***	***3p***	*Δ*
Fe	707	53	654
Co	778	60	718
Ni	853	67	786
Cu	933	75	858
Zn	1022	89	933

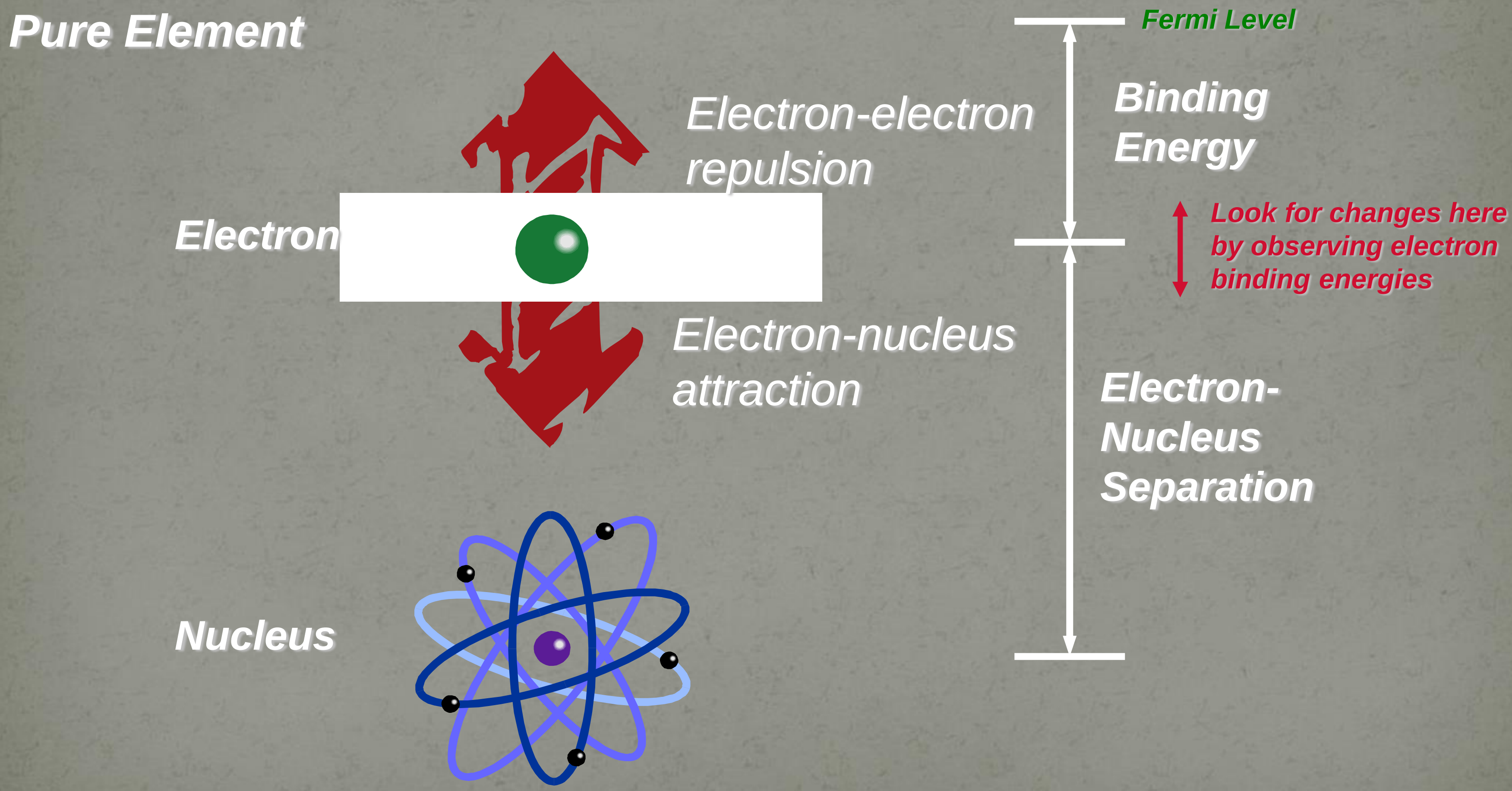
Electron-nucleus attraction helps us identify the elements



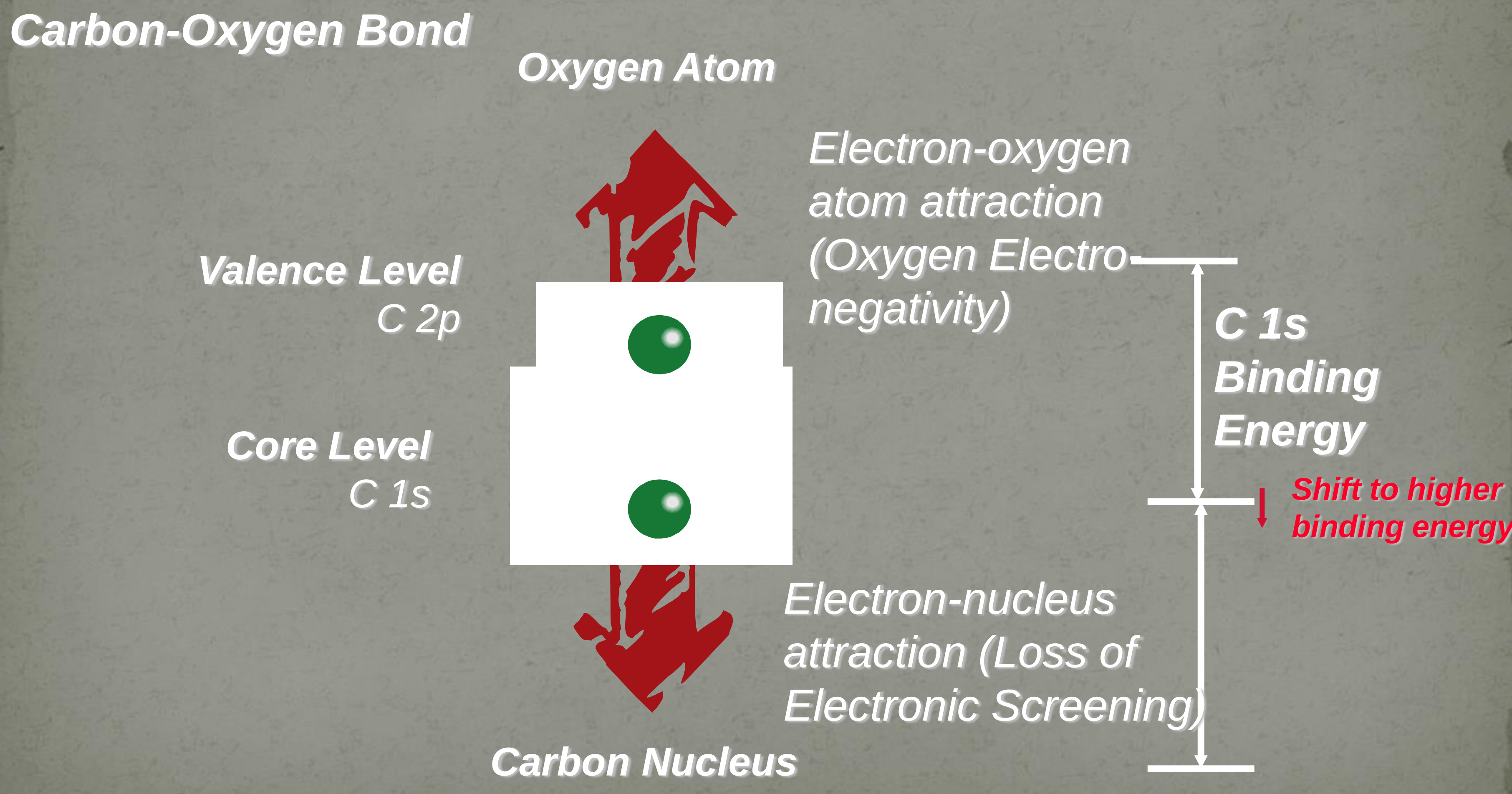
(c) Hf 4f S3
3.0E4
2.0E4
1.0E4
0.0
Hf 4f7/2
(17.02eV)
Hf4+ 4f7/2
(17.16eV)
Hf x+ 4f7/2
(16.54eV)
Hf 4f5/2
(18.67eV)
Hf 4+ 4f5/2
(18.82eV)
Hf x+ 4f5/2
(18.21eV)
Background
12
14
16
18
20
22
24
B.E. (eV)

Where do Binding Energy Shifts Come From?

-or How Can We Identify Elements and Compounds?



Chemical Shifts- Electronegativity Effects



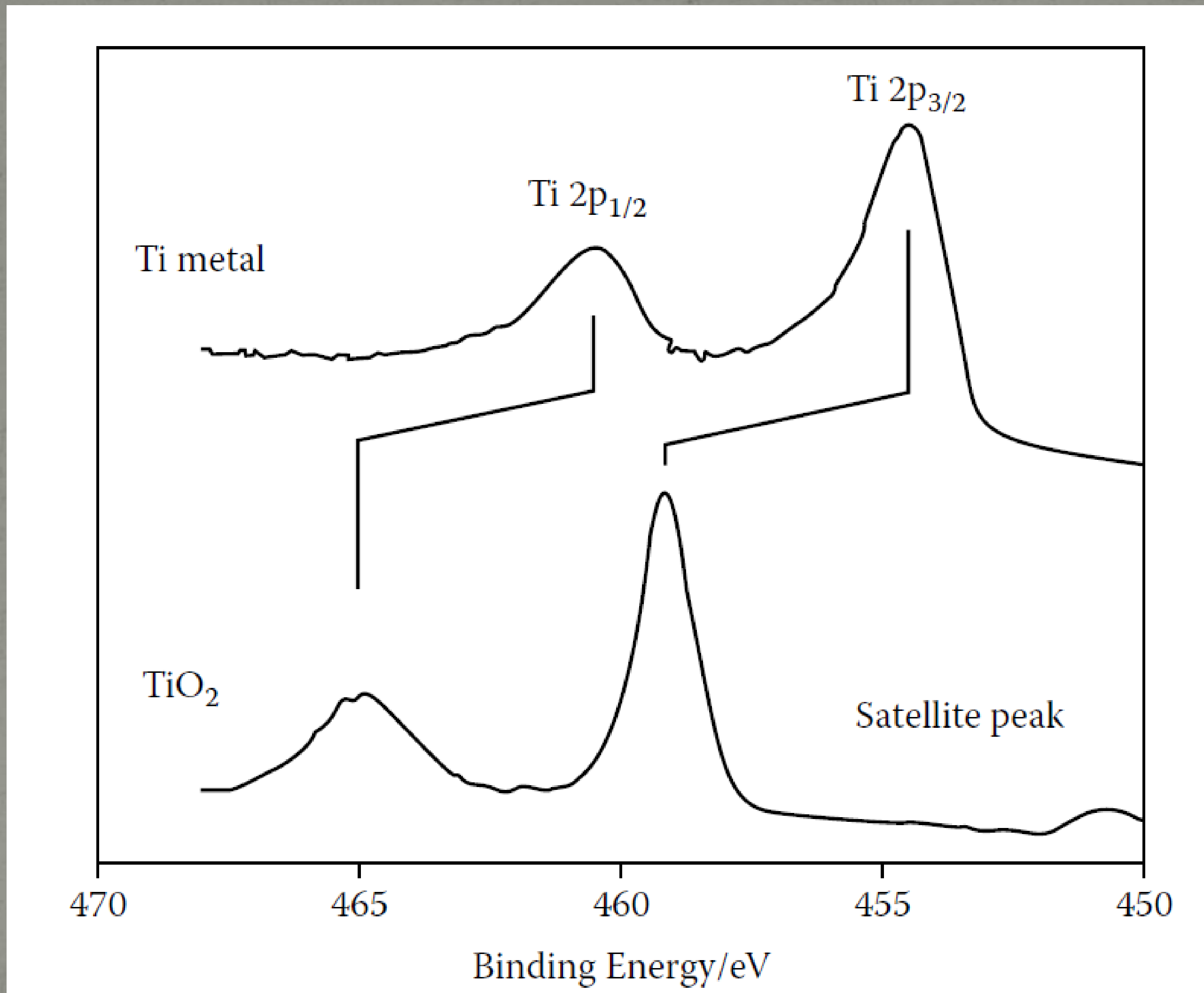
Chemical Shift in XPS

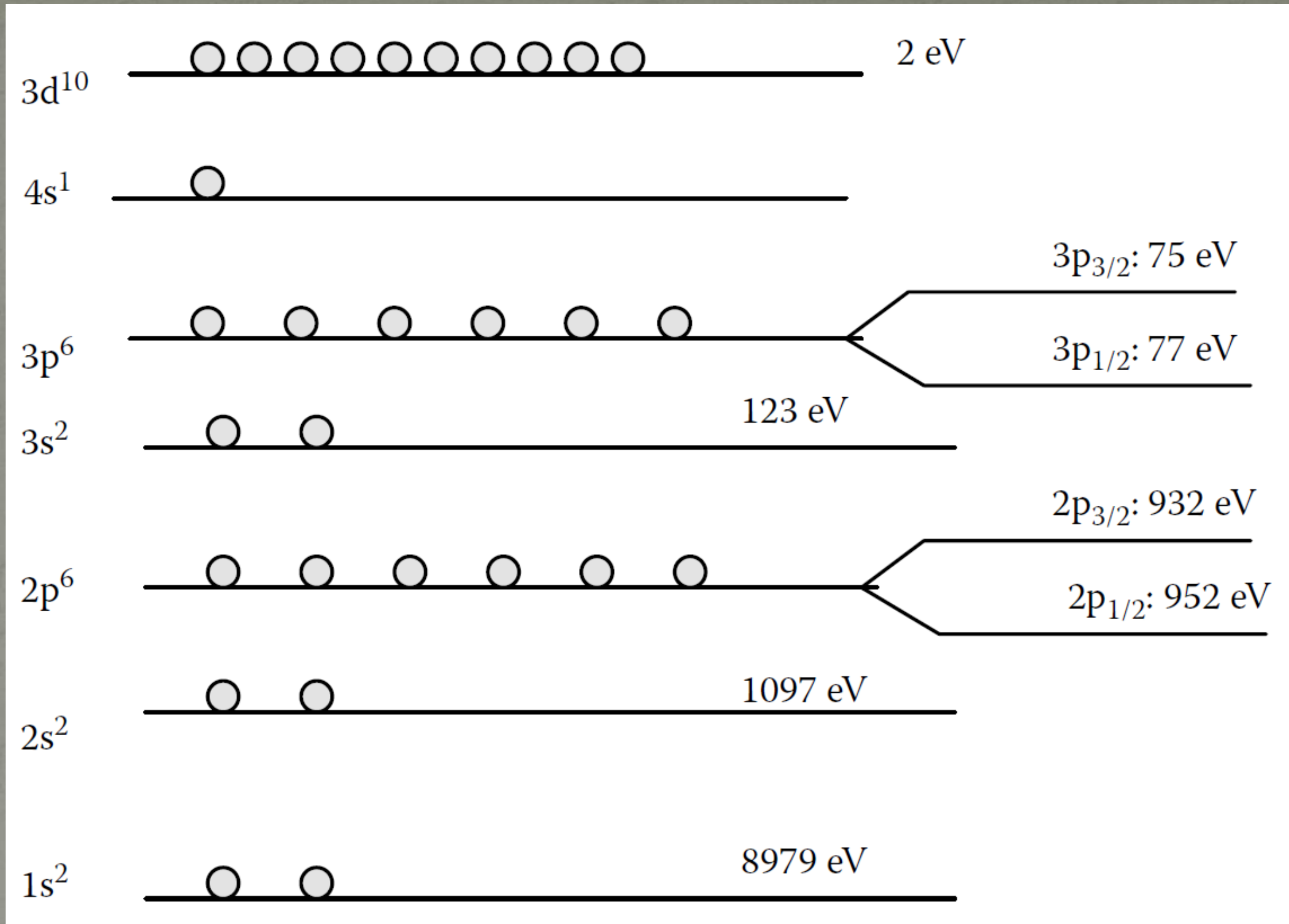
- The initial state is the ground state of the atom before ionization (either by photoionization or primary electron bombardment).

- If the energy of the atom's initial state is changed, for example, by formation of chemical bonds with other atoms or by alteration of the local chemical and physical environment (such as formation of oxide, thus a neutral state becomes a charged state), the energy of electrons in that atom will change.

- The change in energy, is the so-called chemical shift.

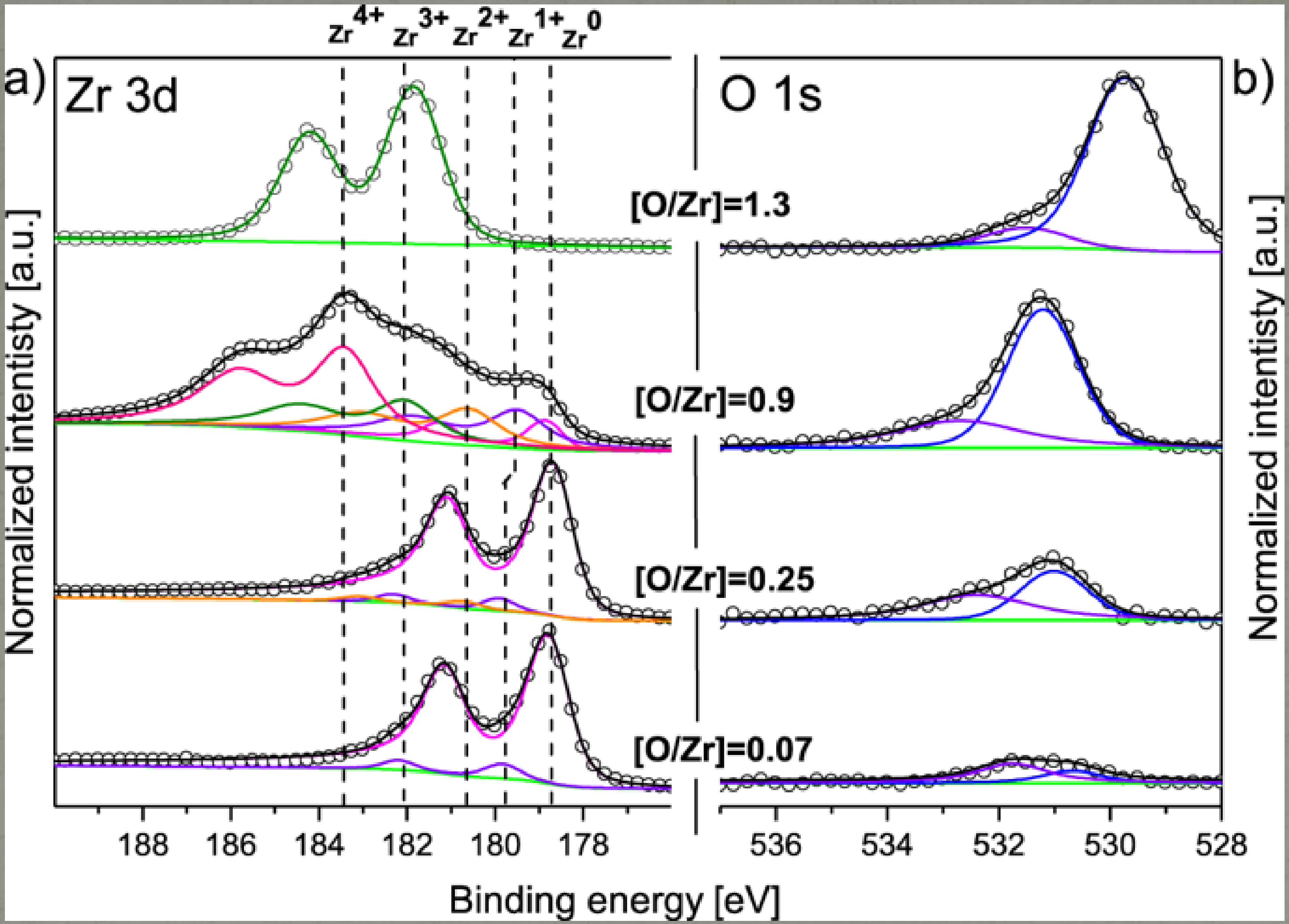
Electronic Effects- Spin-Orbit Coupling





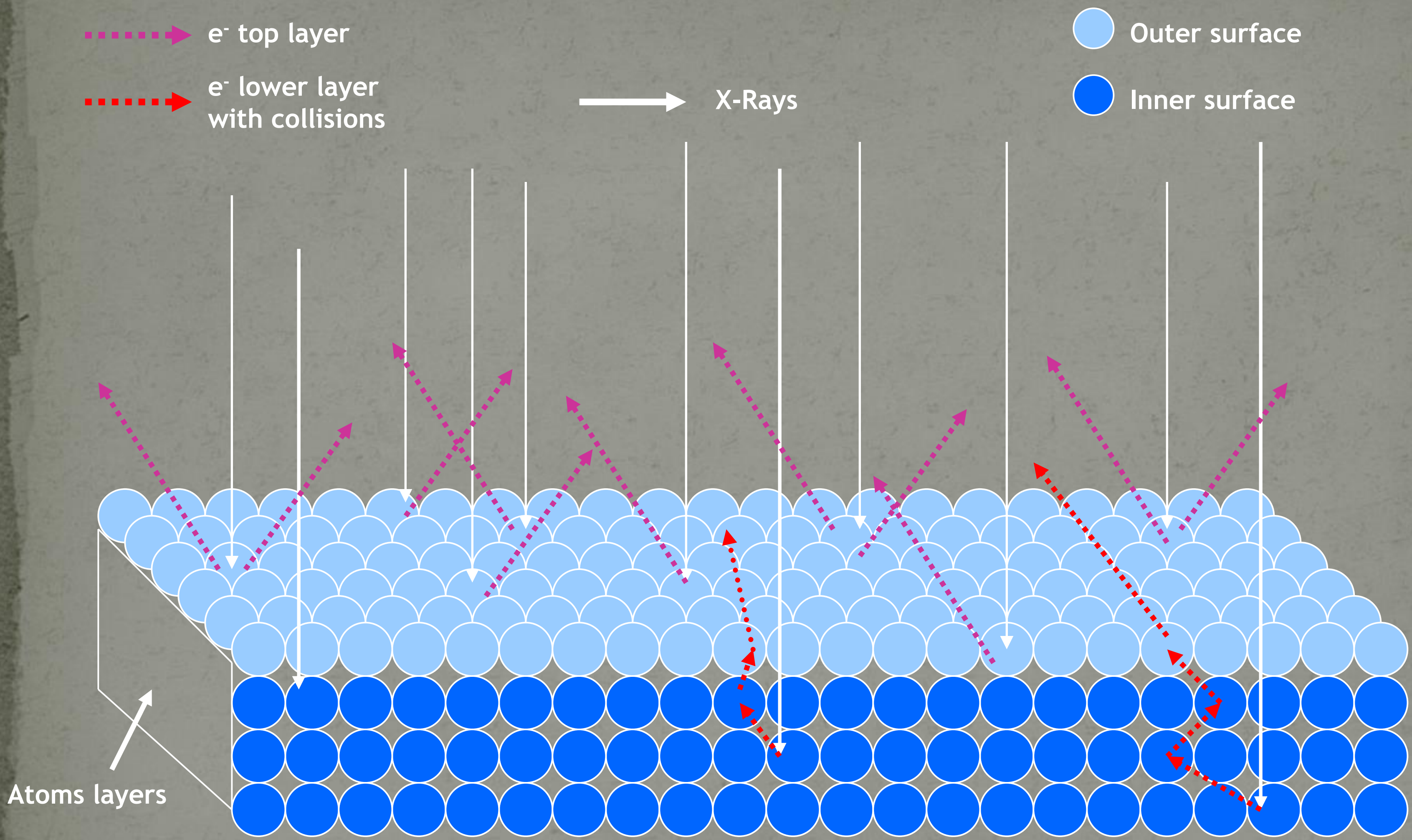
◆Chemical shift refers to a given orbit.

Chemical Shift in XPS



◆ Charged ions with different valences (different environments or coordination) showing different binding energies

X-Rays on the Surface

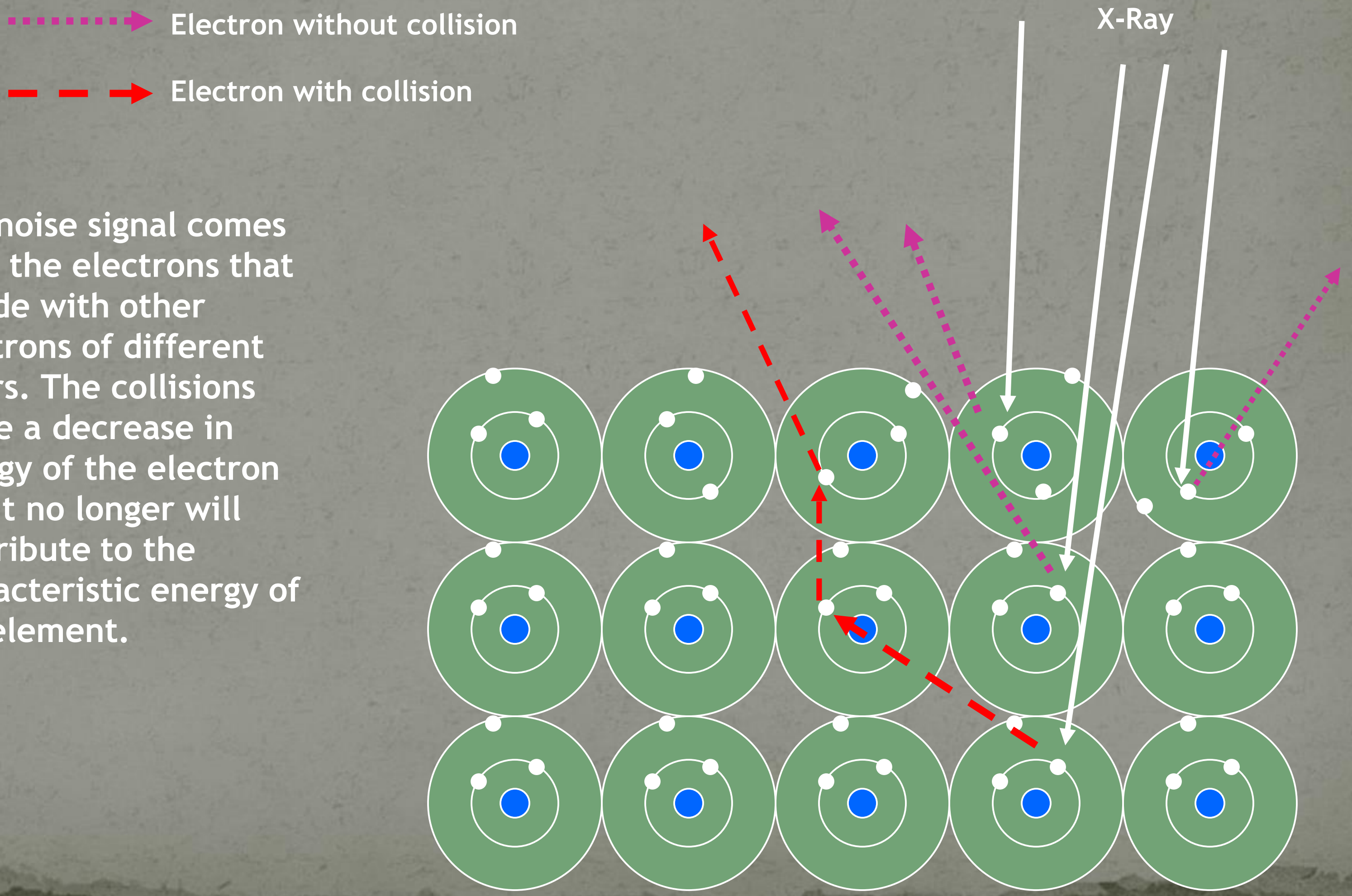


Although x-rays can penetrate much deeper into a material, photoelectrons free of inelastic collision are only from the top 1 to 10 nm of the material; thus, XPS is a very surface-sensitive method for chemical analysis.

X-Rays on the Surface

- The X-Rays will penetrate to the core e^- of the atoms in the sample.
- Some e^-s are going to be released without any problem giving the Kinetic Energies (KE) characteristic of their elements.
- Other e^-s will come from inner layers and collide with other e^-s of upper layers
  - These e- will be lower in lower energy.
  - They will contribute to the noise signal of the spectrum.

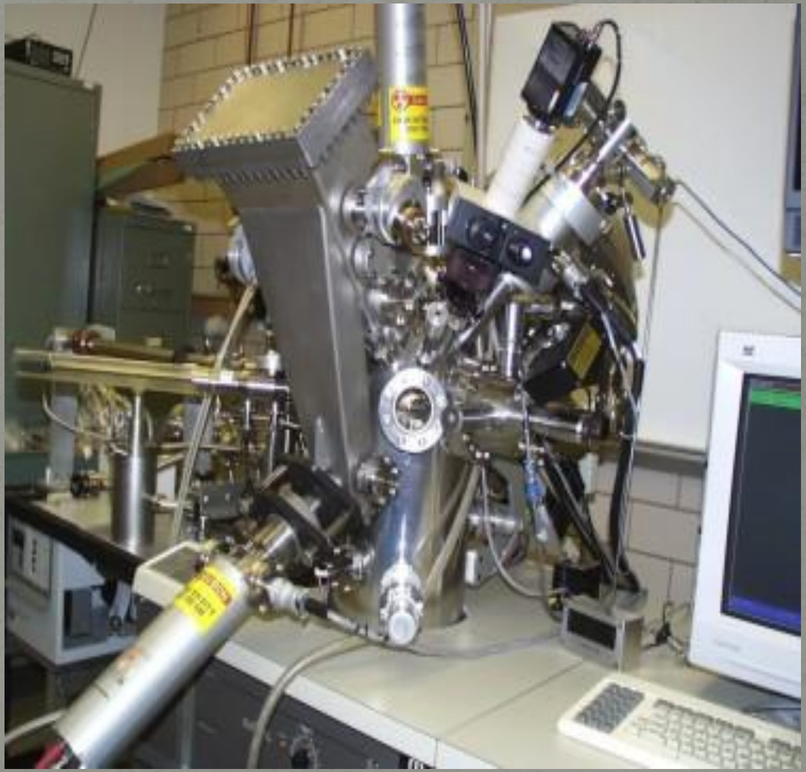
X-Rays and the Electrons



The noise signal comes from the electrons that collide with other electrons of different layers. The collisions cause a decrease in energy of the electron and it no longer will contribute to the characteristic energy of the element.

Instrumentation for X-ray Photoelectron Spectroscopy

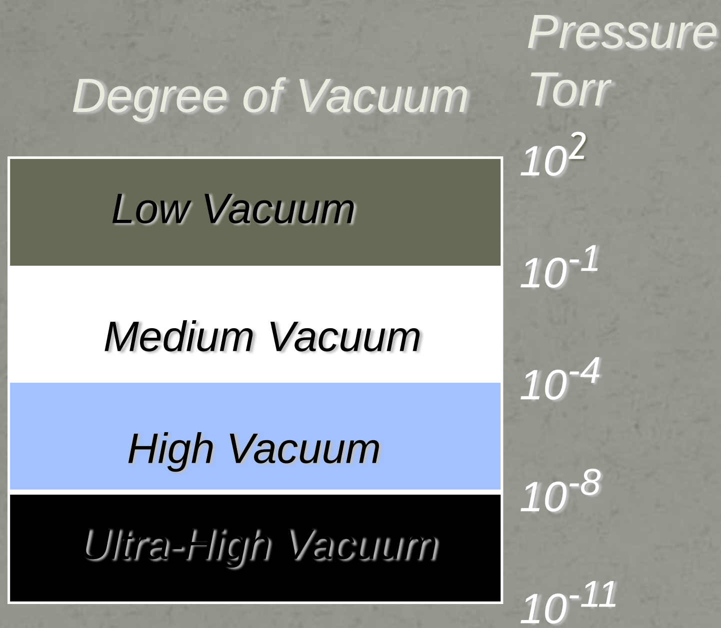
XP SPECTROMETERS



COMPONENTS OF XPS:

- A source of X-rays
- An ultra high vacuum (UHV)
- An electron energy analyzer
- magnetic field shielding
- An electron detector system
- A set of stage manipulators

Why UHV for Surface Analysis?



- *Remove adsorbed gases from the sample.*
- *Eliminate adsorption of contaminants on the sample.*
- *Prevent arcing and high voltage breakdown.*
- *Increase the mean free path for electrons, ions and photons.*

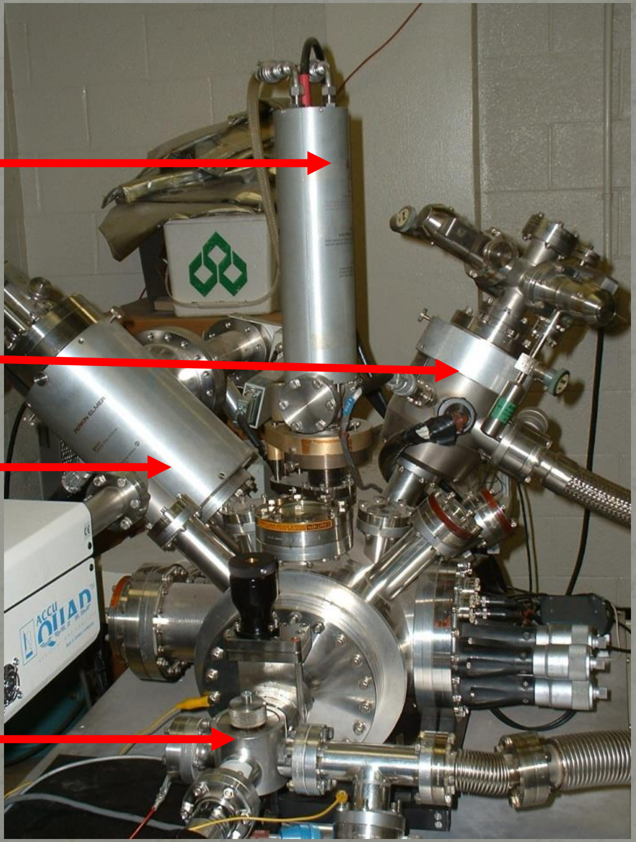
XPS Instrument

X-Ray Source

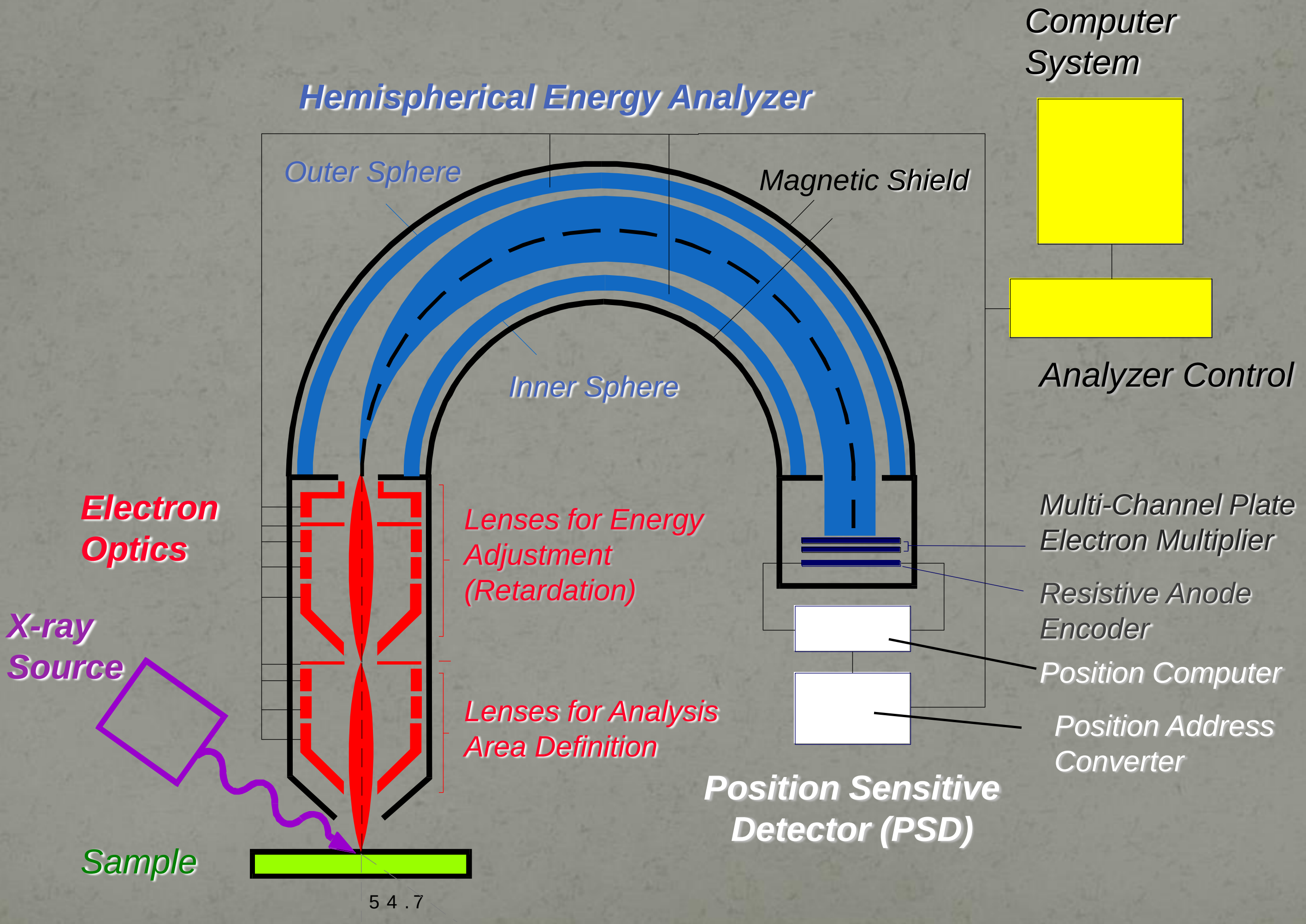
Ion Source

SIMS Analyzer

Sample introduction Chamber



X-ray Photoelectron Spectrometer



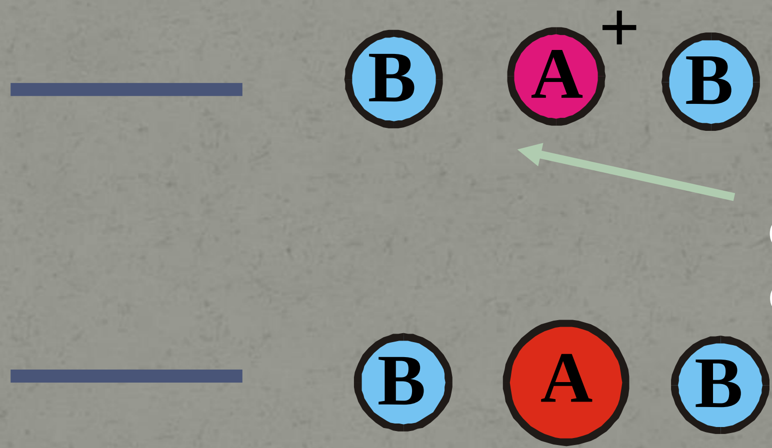
WHY WE USE UHV?

- *Remove adsorbed gases from the sample.*
- *Eliminate adsorption of contaminants on the sample.*
- *Prevent arcing and high voltage breakdown.*
- *Increase the mean free path for electrons, ions and photons.*

Chemical Shift

B.E. = Energy of Final state - Energy of initial state

Final state
(one additional
+ve charge)

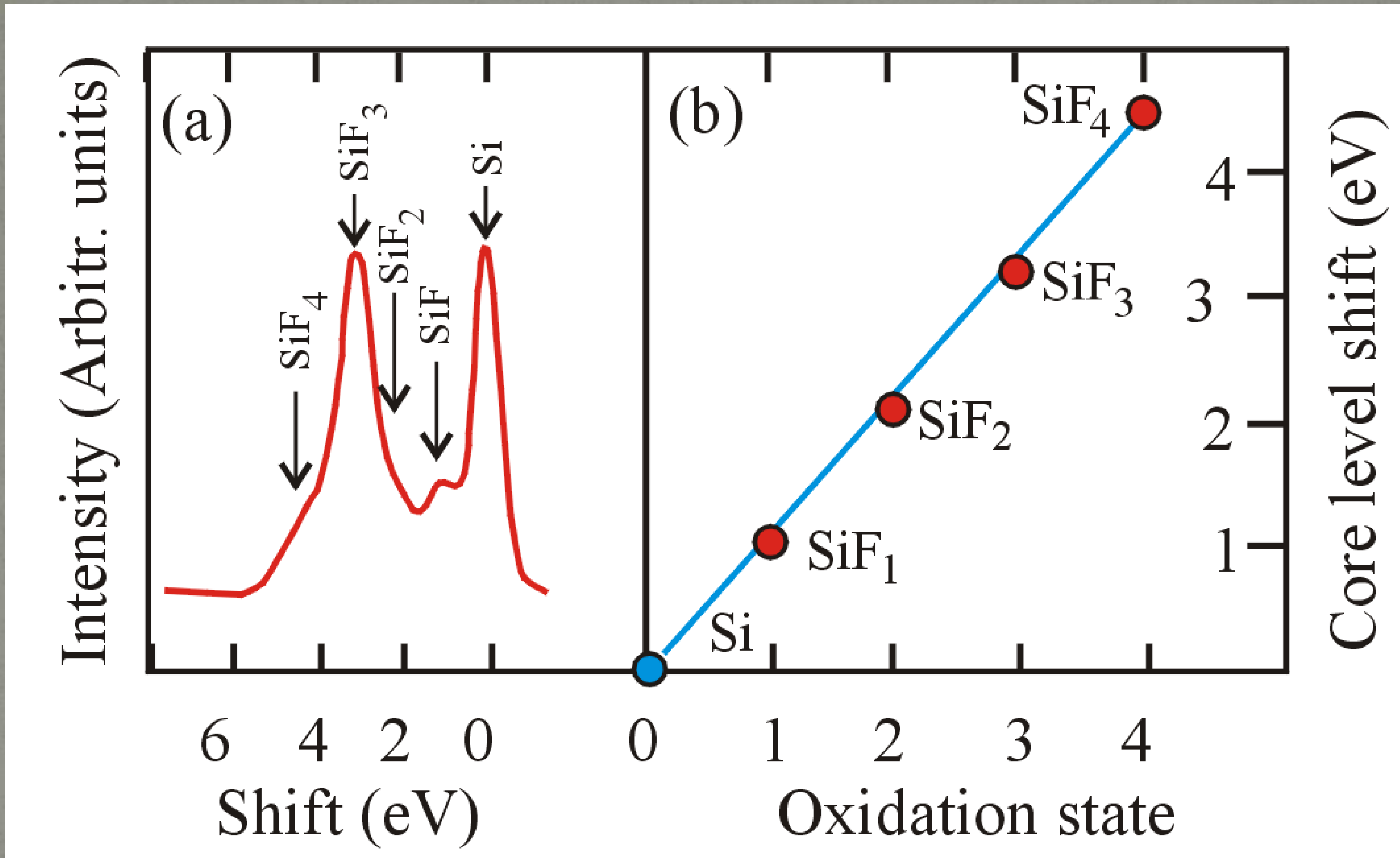


Redistribution
of electron
density

Initial state

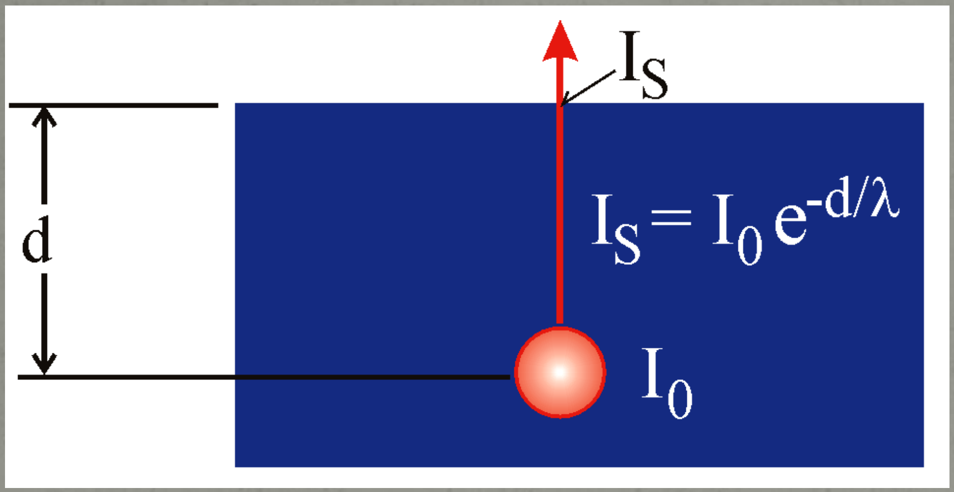
B.E. provides information on chemical environment

Example of Chemical Shift



Analysis Depth

Inelastic mean free path (λ) is the mean distance that an electron travels without energy loss



$$\frac{\int_0^{3\lambda} e^{-\frac{x}{\lambda}} dx}{\int_0^{\infty\lambda} e^{-\frac{x}{\lambda}} dx} = \frac{1 - e^{-3}}{1} = 0.95$$

For XPS, λ is in the range of 0.5 to 3.5 nm

Only the photoelectrons in the near surface region can escape the sample surface with identifiable energy

Measures top 3λ or 5-10 nm

How Does XPS Technology Work?

- A monoenergetic x-ray beam emits photoelectrons from the surface of the sample.
- The x-ray photons penetration about a micrometer of the sample
- The XPS spectrum contains information only about the top 10 - 100 Å of the sample.
- Ultrahigh vacuum environment to eliminate excessive surface contamination.
- Cylindrical Mirror Analyzer (CMA) measures the KE of emitted e^-s.
- The spectrum plotted by the computer from the analyzer signal.
- The binding energies can be determined from the peak positions and the elements present in the sample identified.

which materials are analazıed?

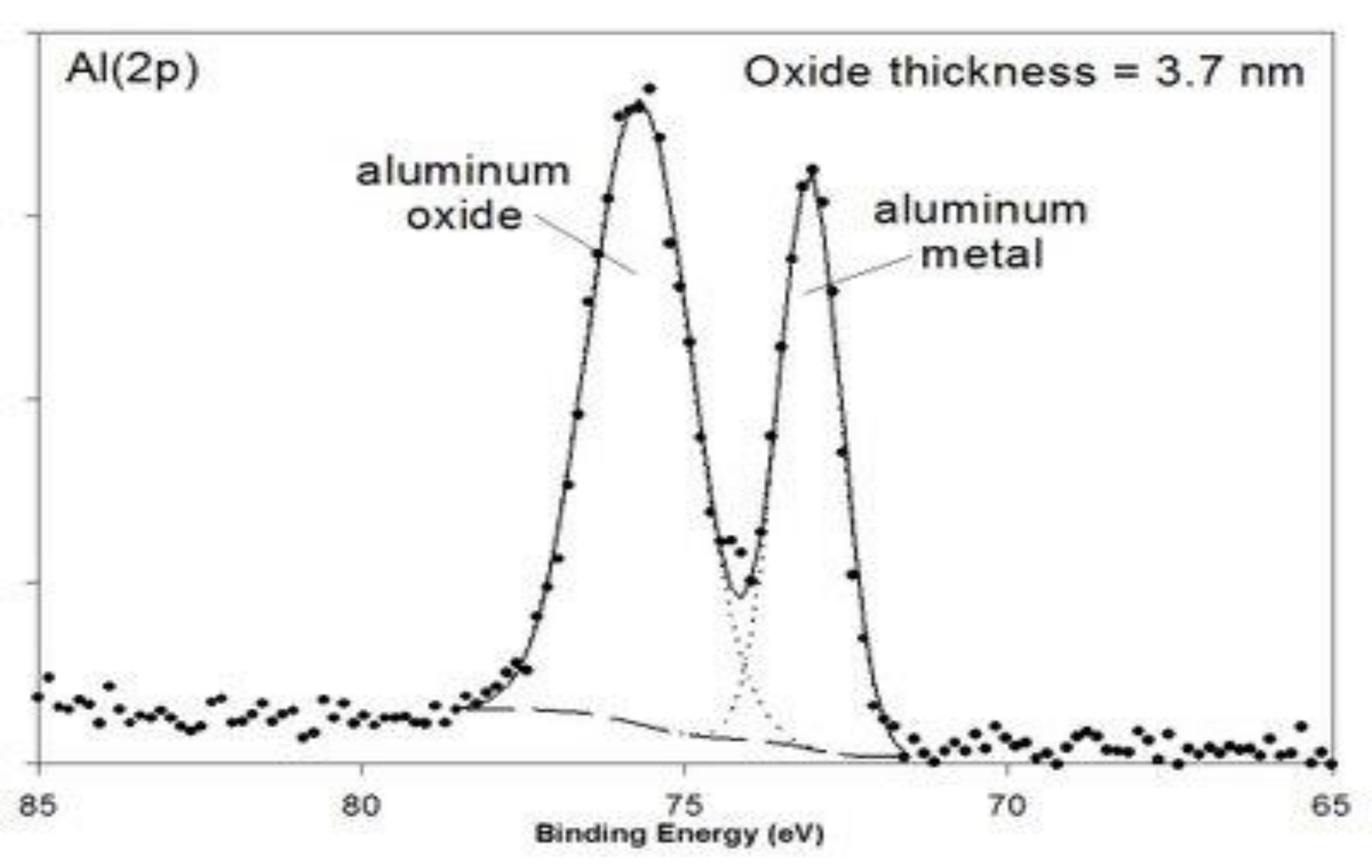
- XPS is routinely used to analyze inorganic compounds,metals,semiconductors,polymers, ceramics,etc.

- Organic chemicals are not routinely analyzed by XPS because they are readily degraded by either the energy of the X-rays or the heat from non-monochromatic X-ray sources

ANALYSIS OF XPS

- XPS detects all elements with (Z) >3. It cannot detect H (Z = 1) or He (Z = 2) because the diameter of these orbitals is so small, reducing the catch probability to almost zero
- Dedection unit: ppt and some conditions ppm
- ppt (parts-per-trillion, 10−12)

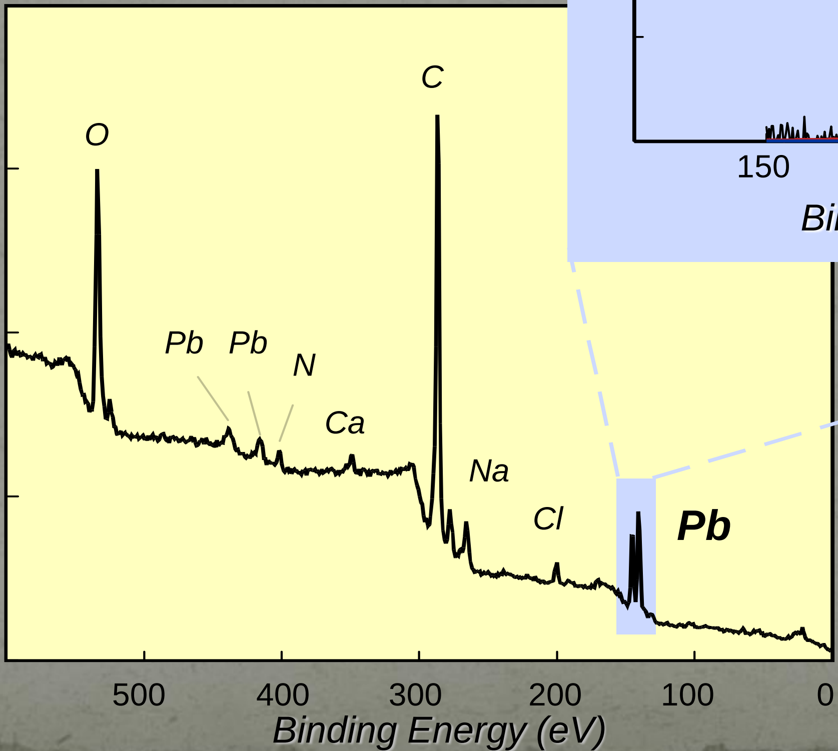
For example:determination of aluminum oxide thickness with xps

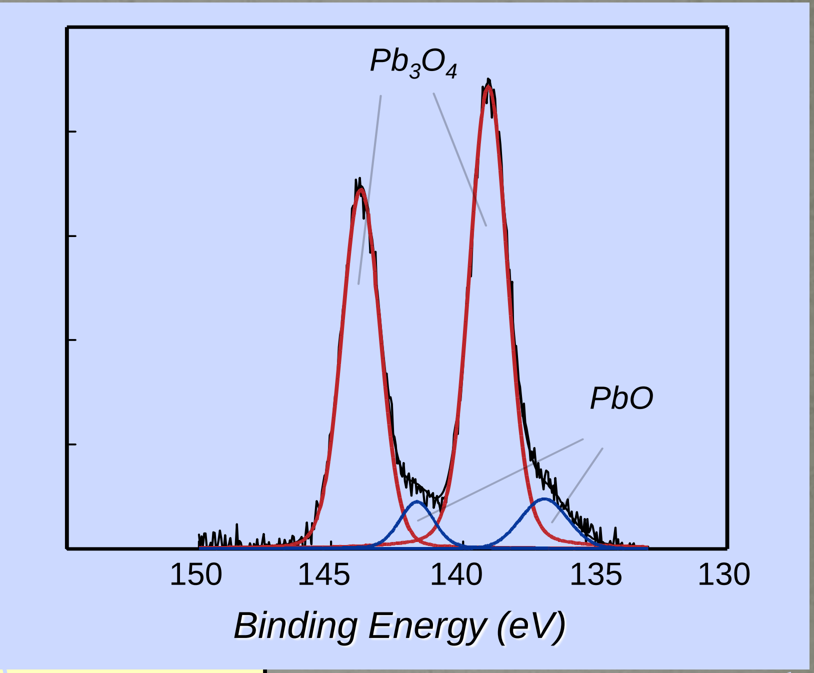


Applications of
X-ray Photoelectron Spectroscopy
(XPS)

XPS Analysis of Pigment from Mummy Artwork

Egyptian Mummy
2nd Century AD
World Heritage Museum
University of Illinois





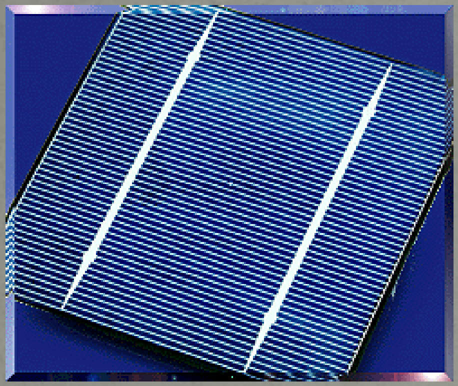
XPS analysis showed that the pigment used on the mummy wrapping was Pb_3O_4 rather than Fe_2O_3

Analysis of Materials for Solar Energy Collection by XPS Depth Profiling-

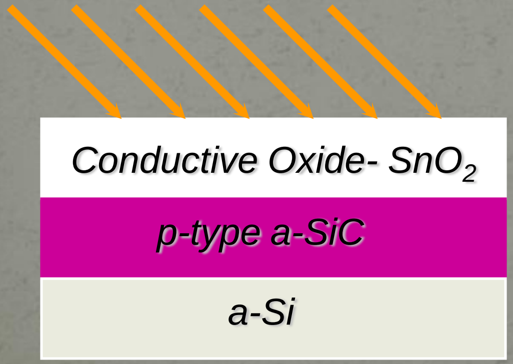
The amorphous-SiC/SnO_2 Interface

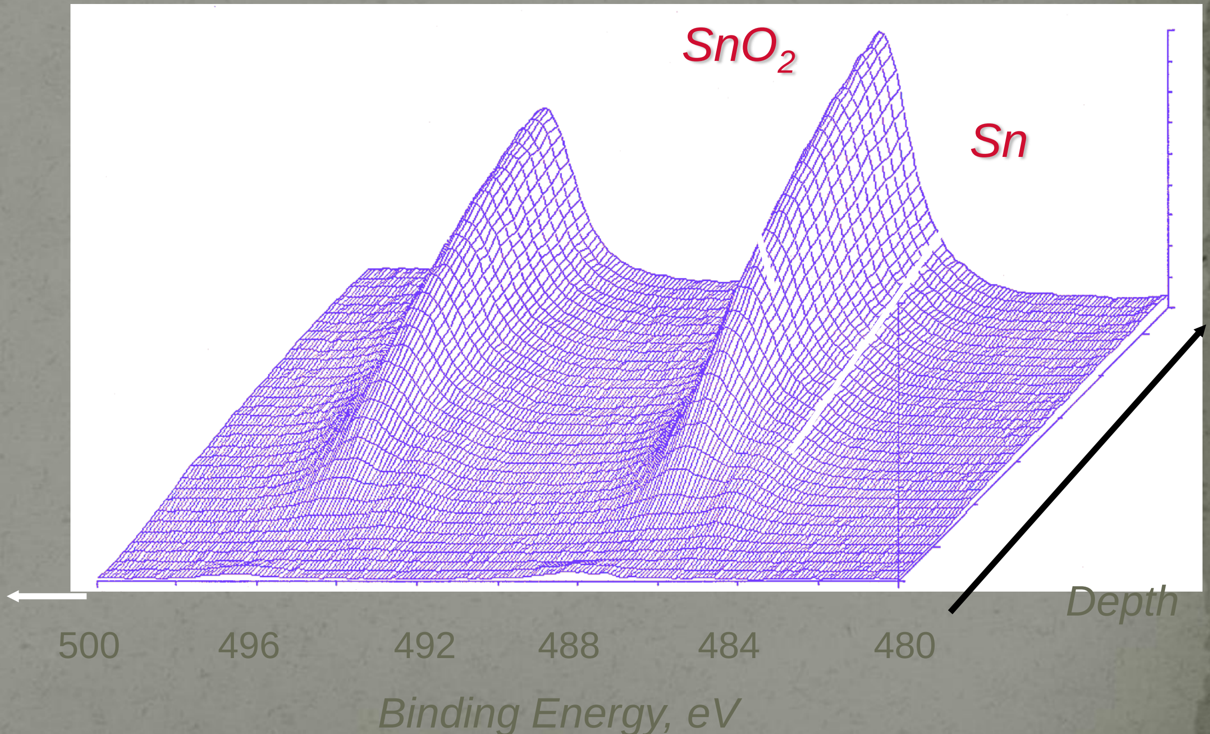
The profile indicates a reduction of the SnO_2 occurred at the interface during deposition. Such a reduction would effect the collector's efficiency.

Photo-voltaic Collector

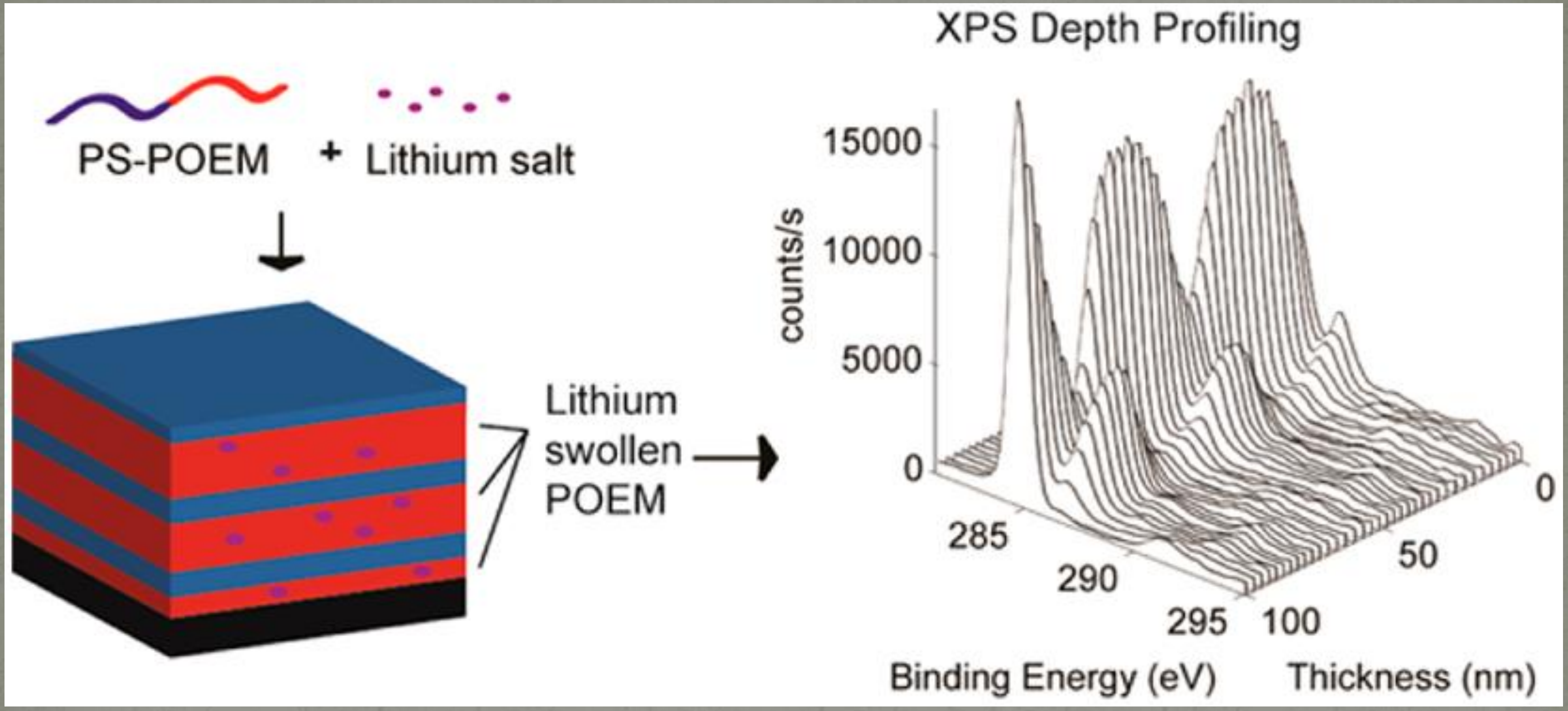


Solar Energy

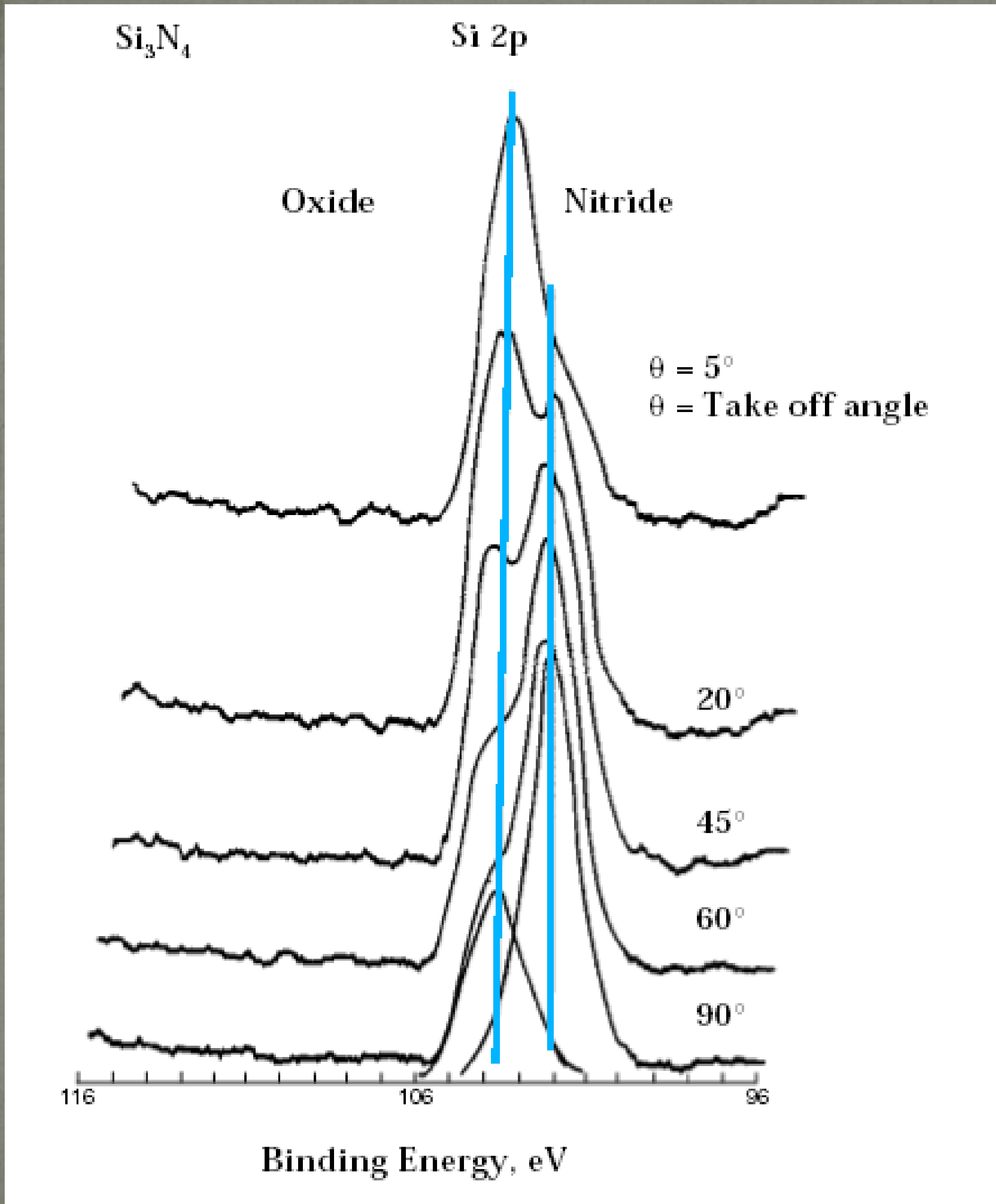




Data courtesy A. Nurrudin and J. Abelson, University of Illinois



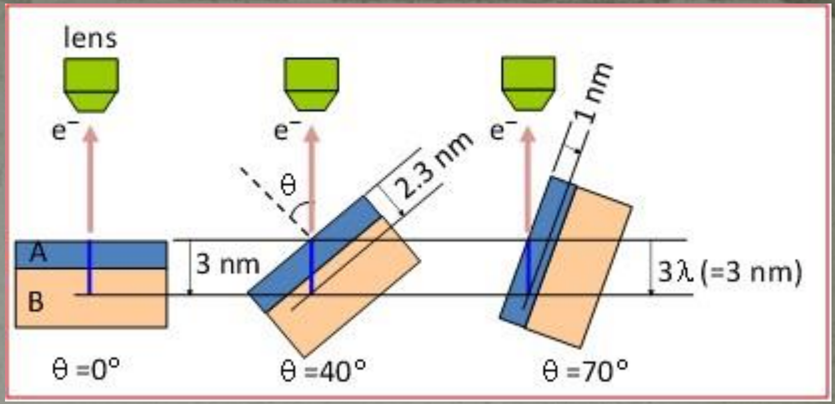
PS-POEM
+ Lithium salt
Lithium
swollen
POEM
XPS Depth Profiling
counts/s
15000
10000
5000
0
285
290
295
100
50
0
Binding Energy (eV)
Thickness (nm)



Angle-resolved XPS analysis

Oxide on silicon nitride surface

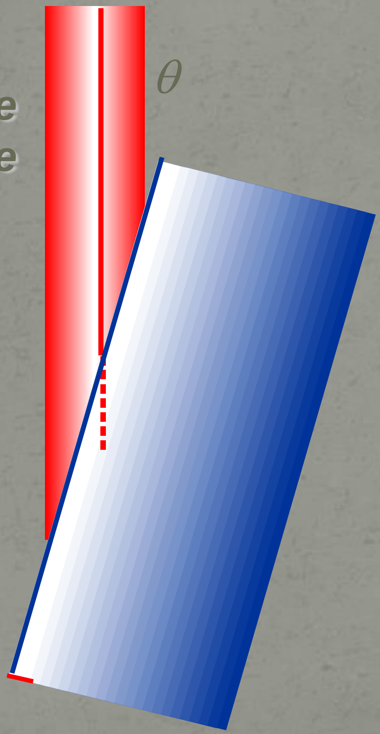
Angle-resolved XPS



θ =15 °

θ

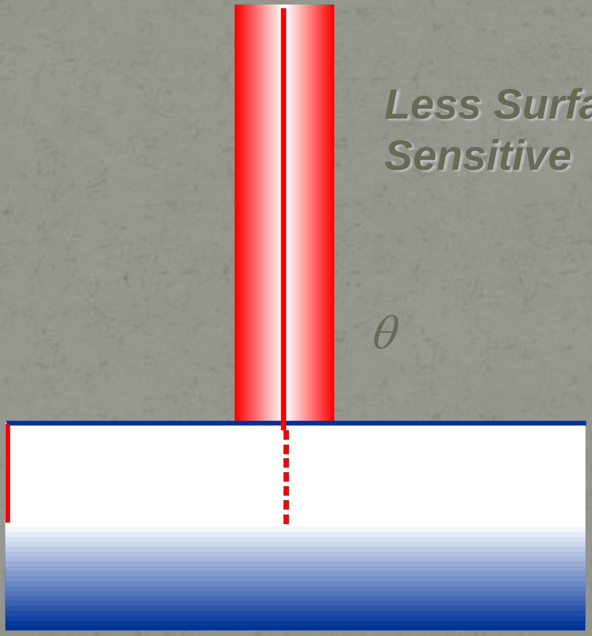
More Surface Sensitive



θ = 90 °

Less Surface Sensitive

θ



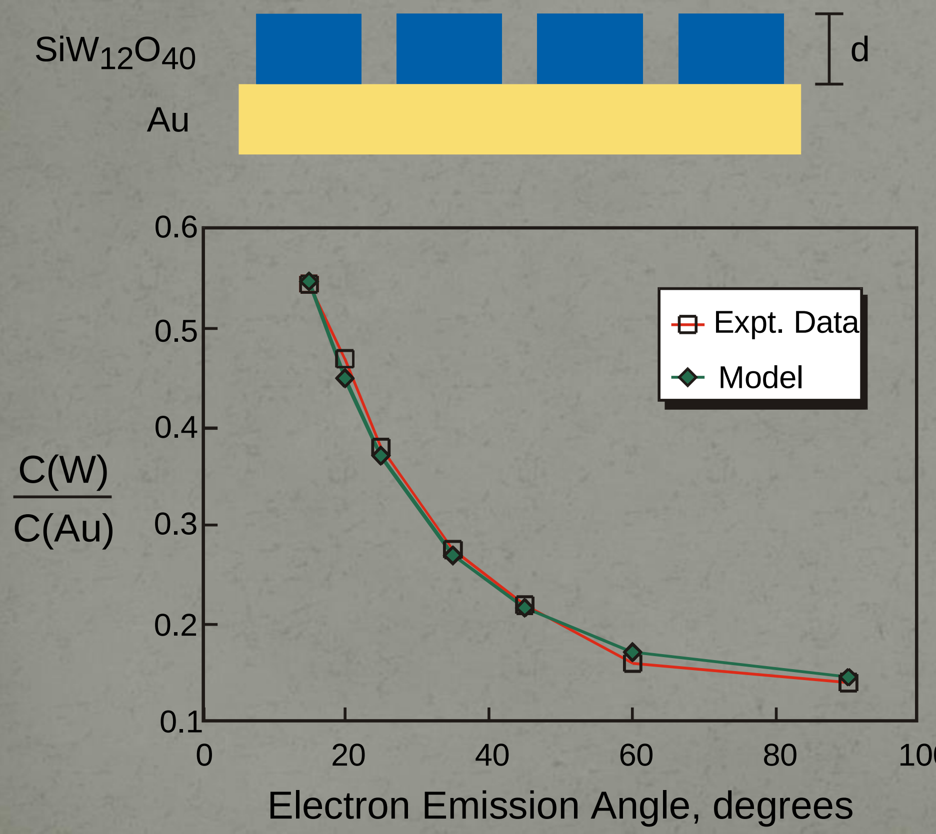
Information depth = dsinθ

d = Escape depth ~ 3 λ

θ = Emission angle relative to surface

λ = Inelastic Mean Free Path

Angle-resolved XPS Analysis of Self-Assembling Monolayers



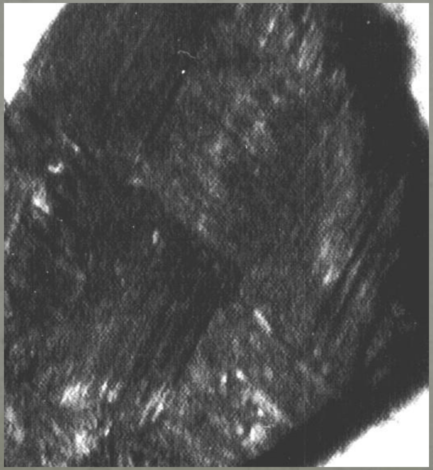
Angle Resolved XPS Can Determine

- *Over-layer Thickness*
- *Over-layer Coverage*

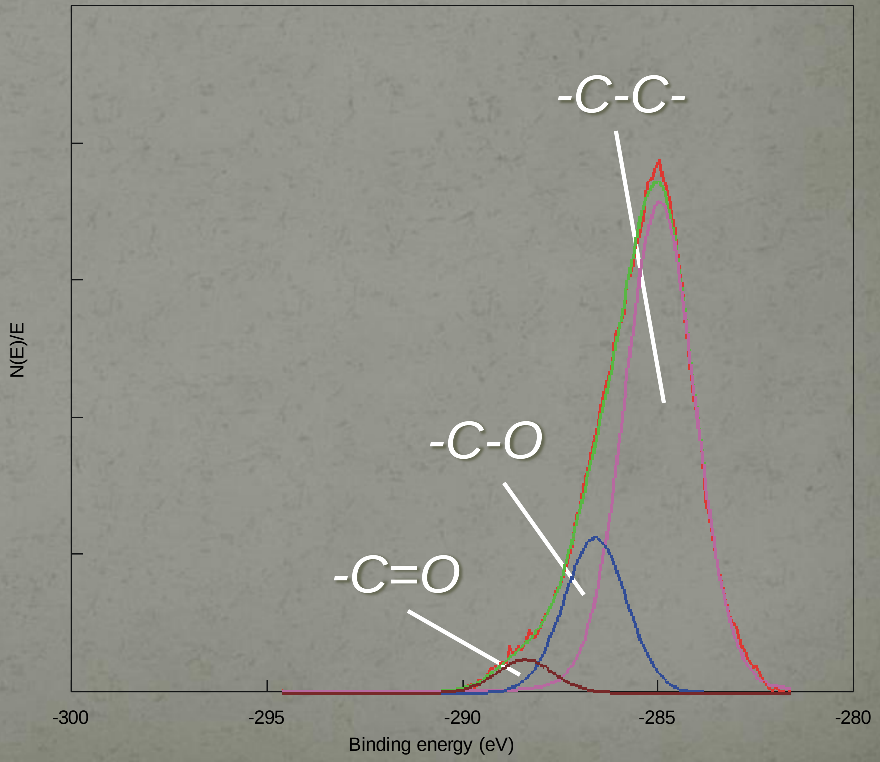
Data courtesy L. Ge, R. Haasch and A. Gewirth, University of Illinois

Analysis of Carbon Fiber- Polymer Composite Material by XPS

XPS analysis identifies the functional groups present on composite surface. Chemical nature of fiber-polymer interface will influence its properties.



Woven carbon fiber composite



Chemical Shifts- Electronegativity Effects

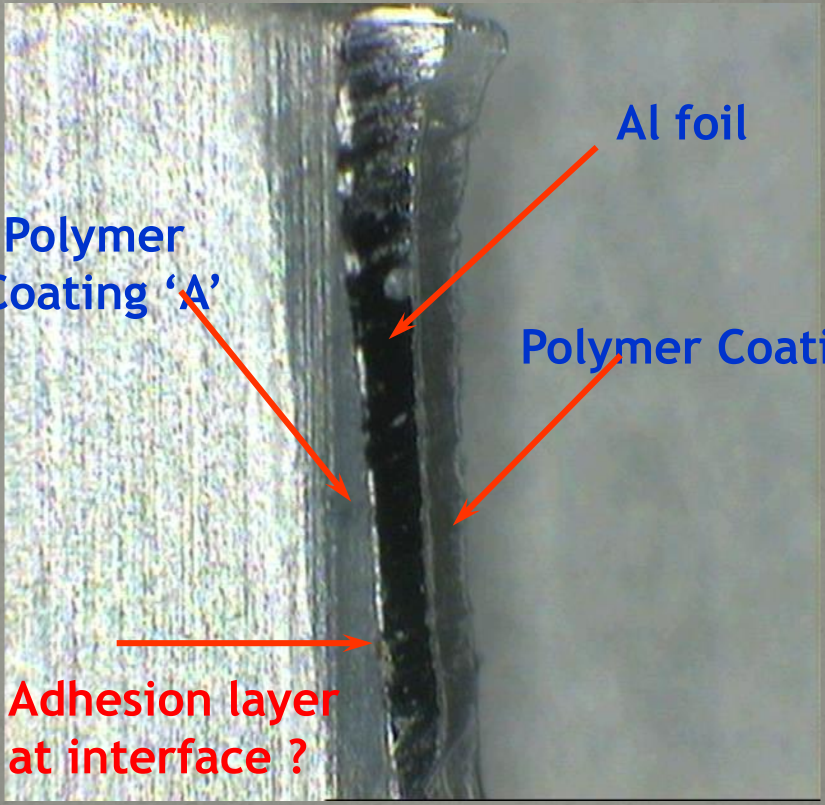
Functional Group		Binding Energy (eV)
hydrocarbon	C-H, C-C	285.0
amine	C-N	286.0
alcohol, ether	C-O-H, C-O-C	286.5
Cl bound to C	C-Cl	286.5
F bound to C	C-F	287.8
carbonyl	C=O	288.0

Multi-layered Drug Package

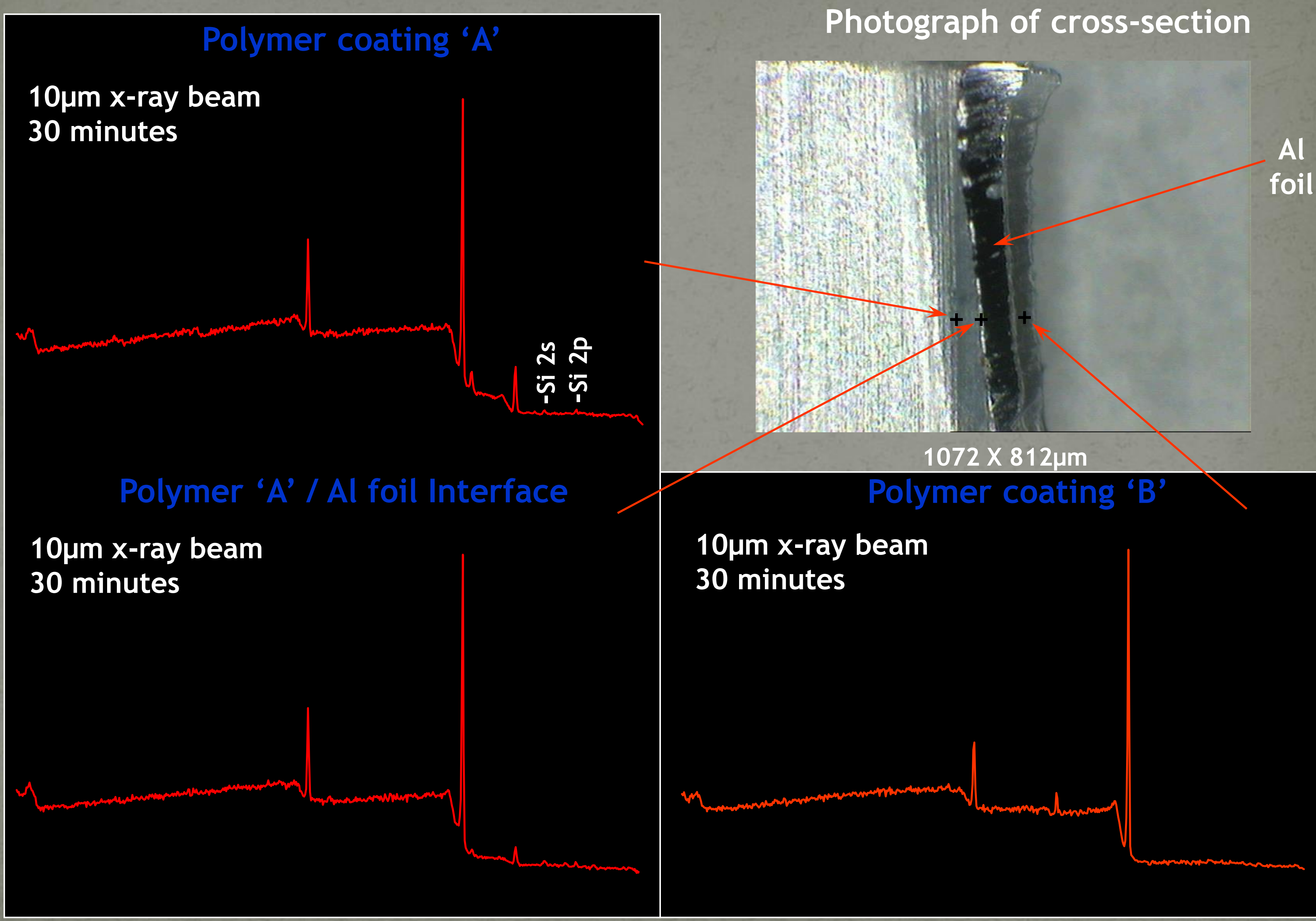
Optical photograph of encapsulated drug tablets

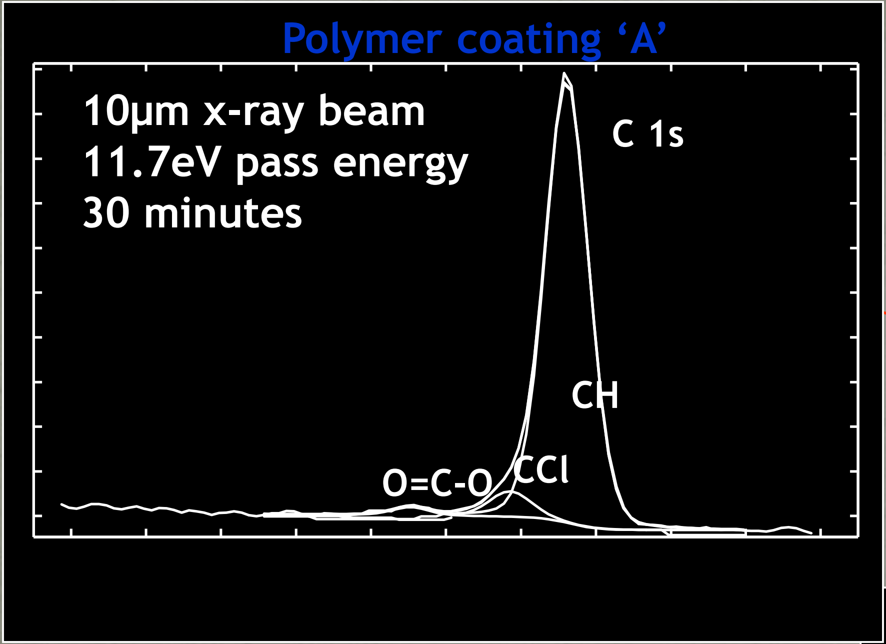
100 X 100mm

SPS Photograph
Cross-section of Drug Package

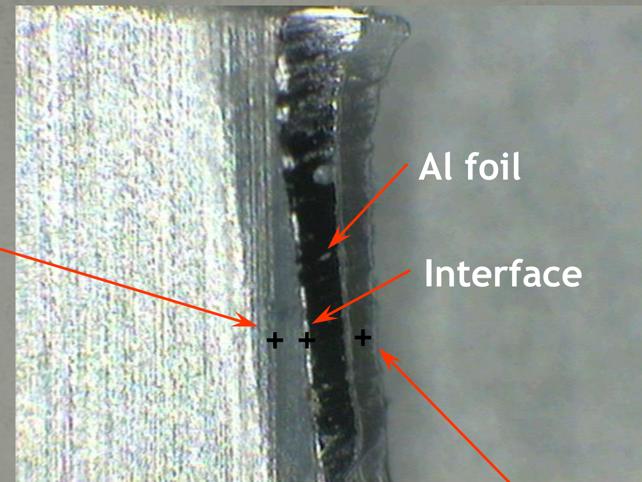


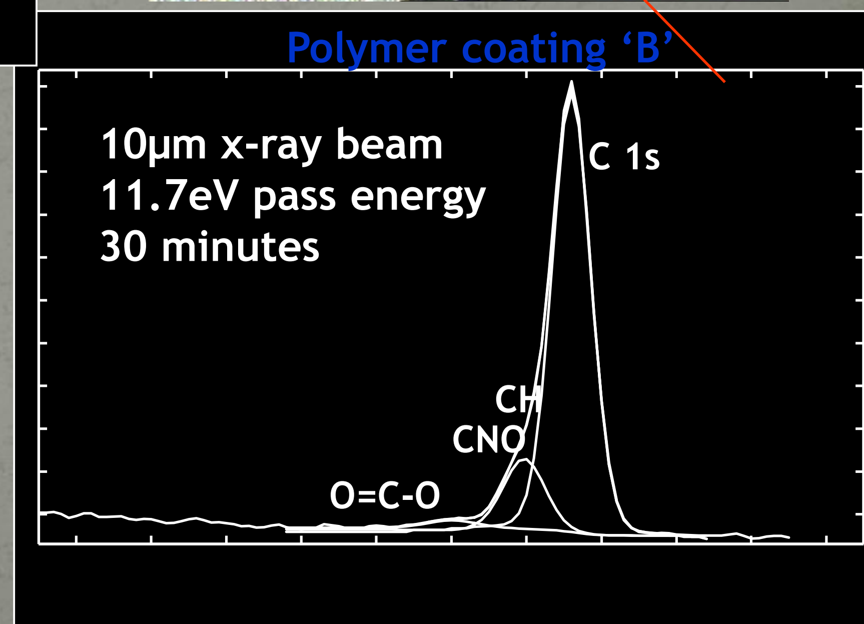
1072 X 812μm


Polymer coating 'A'
10µm x-ray beam
30 minutes
-Si 2s
-Si 2p
Photograph of cross-section
Al
foil
1072 X 812µm
Polymer 'A' / Al foil Interface
10µm x-ray beam
30 minutes
Polymer coating 'B'
10µm x-ray beam
30 minutes



Photograph (1072 X 812um)





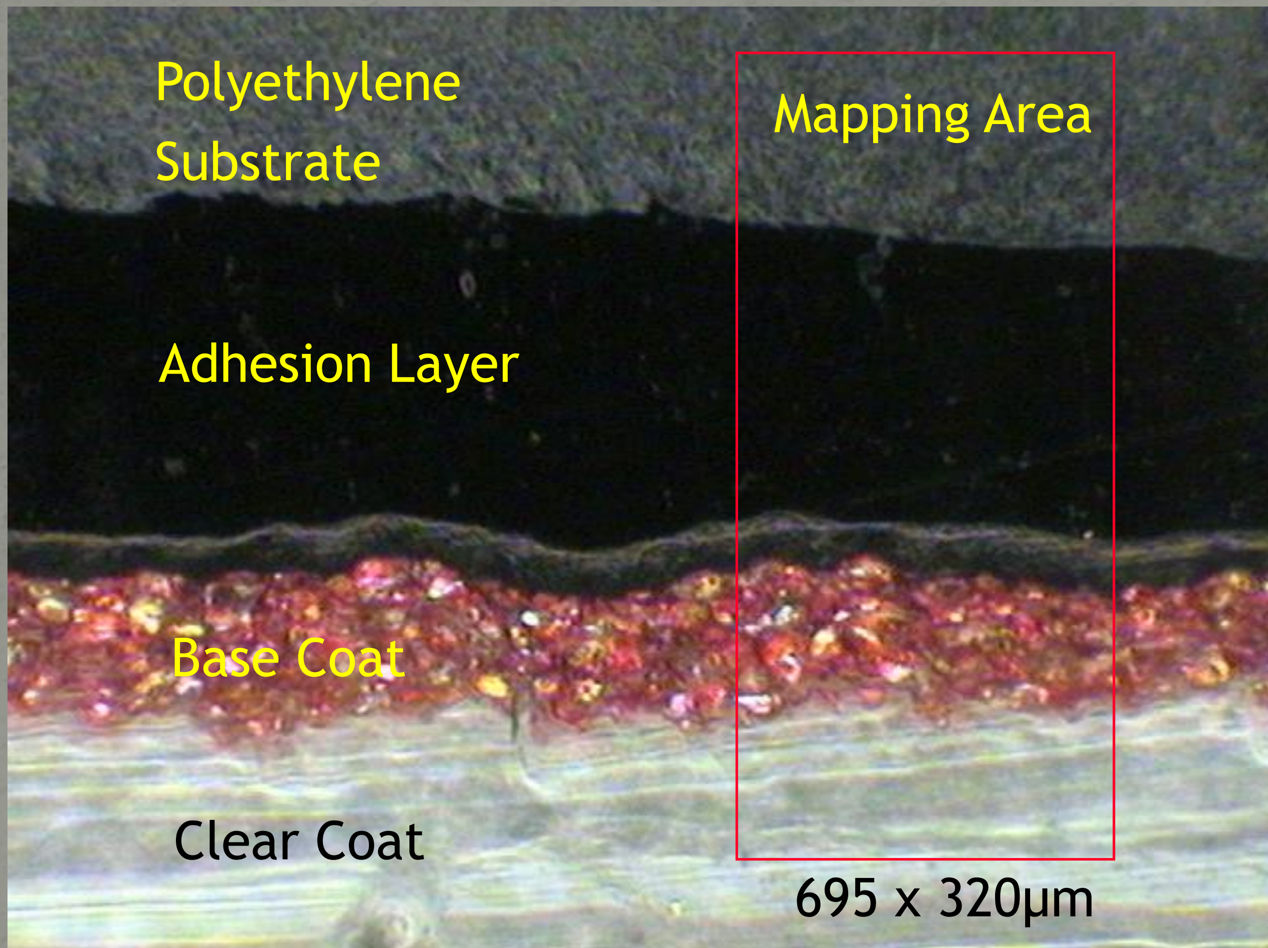
Atomic Concentration (%)

Area	C	O	N	Si
A	82.6	12.2	----	0.7
Interface	83.2	12.2	----	1.3
B	85.9	9.8	4.3	----

A silicon (Si) rich layer is present at the interface

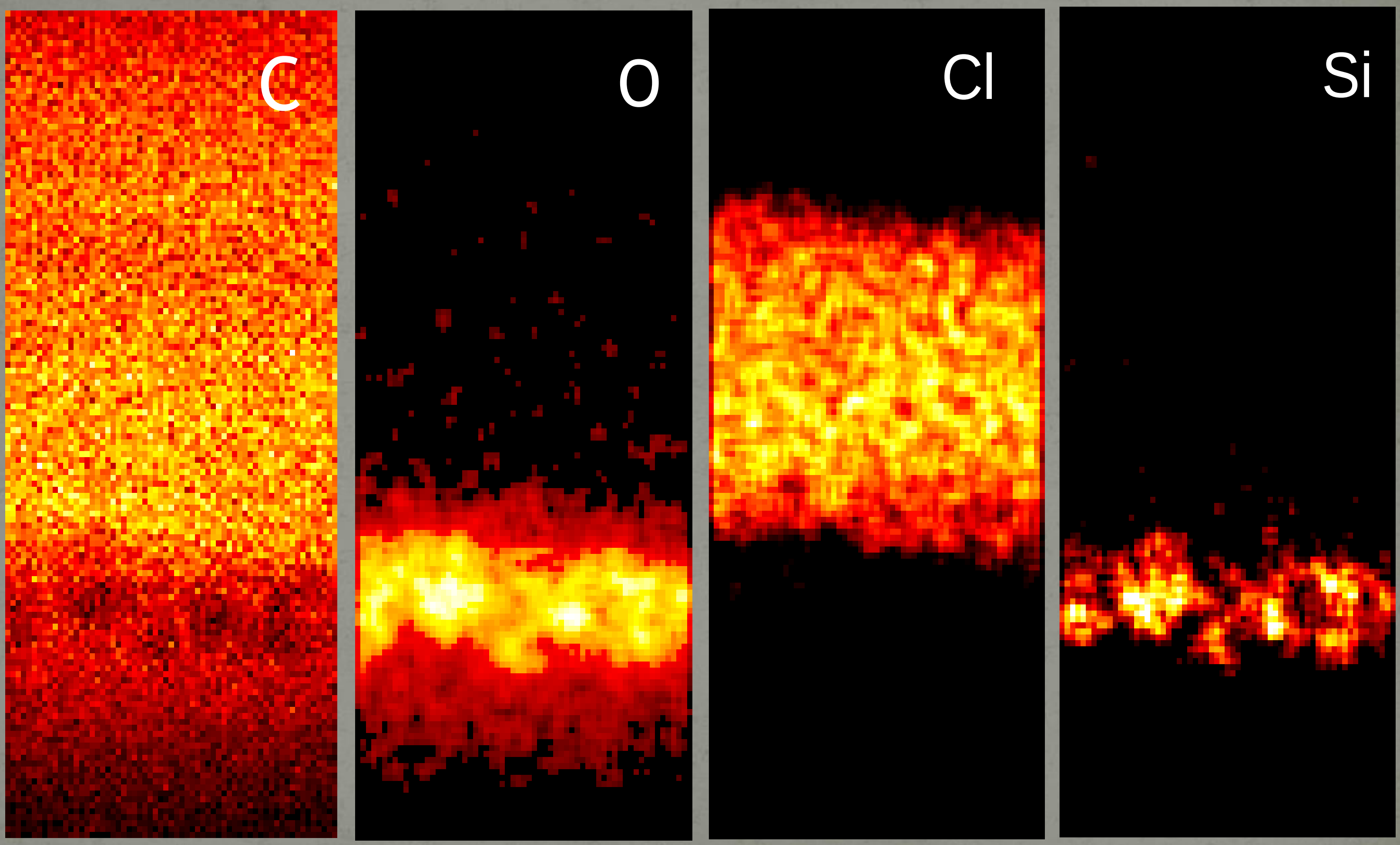
XPS study of paint

Paint Cross Section



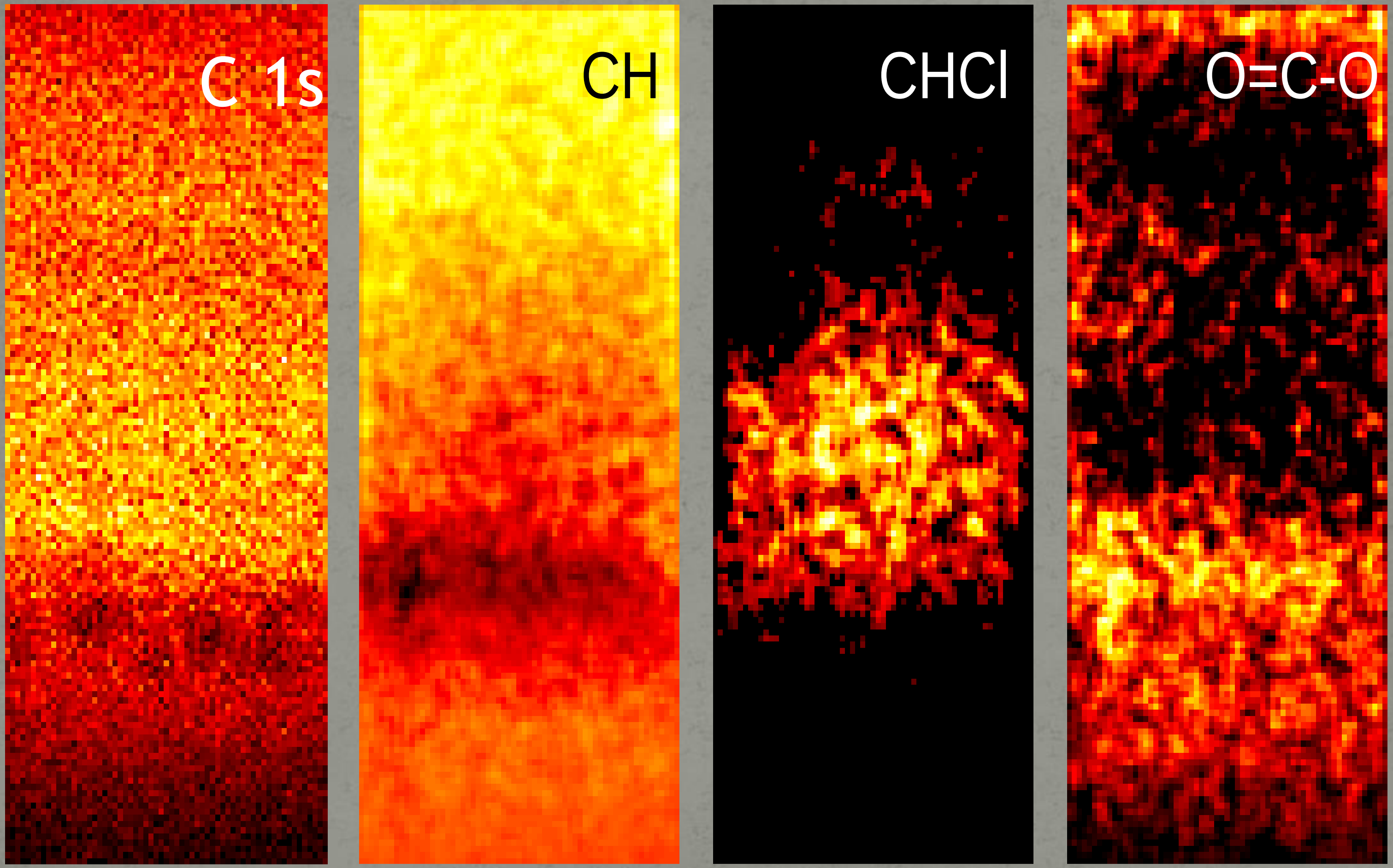
1072 x 812mm

Elemental ESCA Maps using C 1s, O 1s, Cl 2p, and Si 2p signals



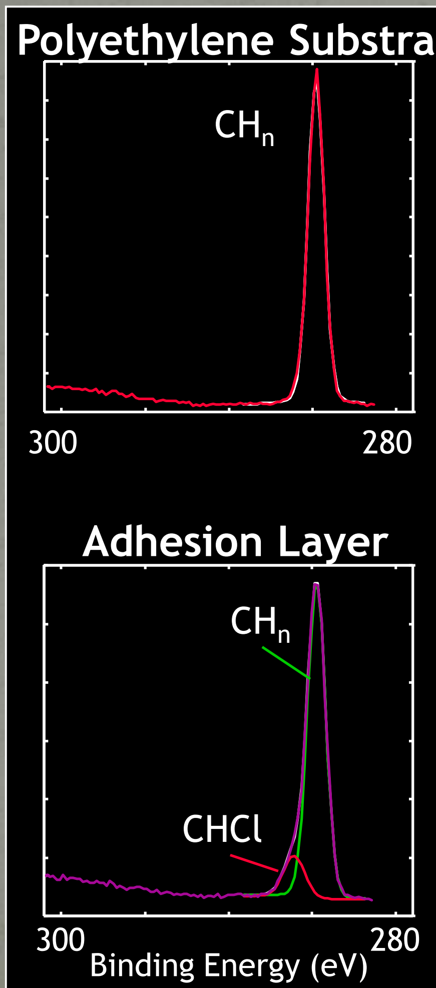
695 x 320mm

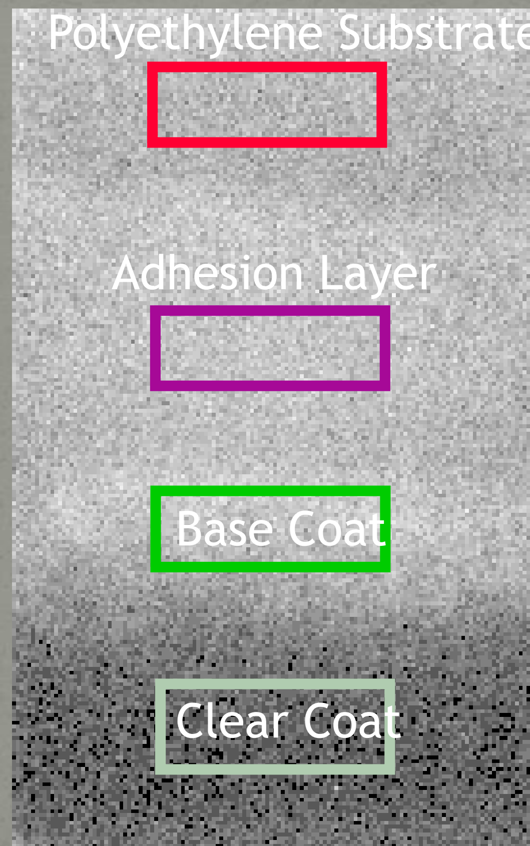
C 1s Chemical State Maps



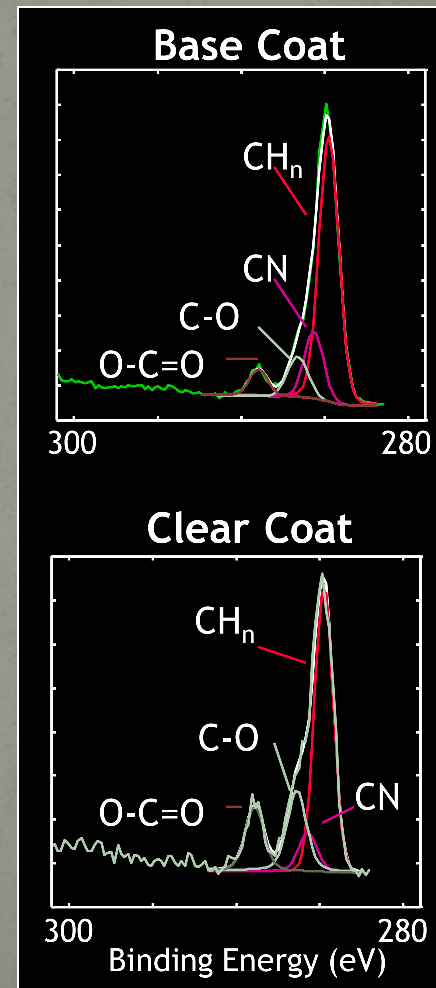
695 x 320mm

Small Area Spectroscopy
High resolution C 1s spectra from each layer





800 x 500µm



Quantitative Analysis

Atomic Concentration* (%)

Analysis Area	C	O	N	Cl	Si	Al
Substrate	100.0	---	---	---	---	---
Adhesion Layer	90.0	---	---	10.0	---	---
Base Coat	72.0	16.4	3.5	3.3	2.6	2.2
Clear Coat	70.6	22.2	7.2	---	---	---

*excluding H

conclusion

XPS devices is also named ESCA(Electron Spectroscopy Chemical Analysis)It works base on the photoelectron effect .

XPS is primarly used for chemical analysis as determining thicknesses and emprical formulas of different elements and,binding energies,densities of electronic states.

XPS is used from starting Li(A>=3) because of smaller orbital's radius and analysis of organic compounds cannot be done with XPS because of degredation with radiation.

XPS is used also determination of others like; metals,ceramics,semiconductors,papers,etc.

References:

- Siegbahn, K.; Edvarson, K. I. Al (1956). "β-Ray spectroscopy in the precision range of 1 : 1e6". *Nuclear Physics*
- Turner, D. W.; Jobory, M. I. Al (1962). "Determination of Ionization Potentials by Photoelectron Energy Measurement".
- *journals.tubitak.gov.tr*
- *nanohub.org*
- srdata.nist.gov
- www.eaglabs.com

  www.files.chem.vt.edu
- Bio interface.org
- *www.*spectroscopy*now.com*
- *www.surfaceanalysis.org*
- www.csma.ltd.uk